Model Structure (26.2_FAA):

This is a Fleets−as−Areas model (do_spatial == 1).
There are 2 sexes, 2 fishing fleets, 5 survey fleets, and 1 regions being modeled.
The model has 67 years, starting in 1960 and ending in 2026.
There are 30 ages, where recruitment occurs at age−2 and the plus group occurs at age−31.
There are 30 length bins, which start at 41cm and end at 99cm.

Growth Estimation:

This model estimates new growth curves for the post−1996 time block (upd_growth == 1, use_new_bio == 2) and uses the 2021 SAFE growth estimates for the pre−1996 time block (use_hist_grwth_blk == 1, do_rec_growth == 0). A new weight curve is estimated with a single block (do_rec_WAA == 0).
For fishery composition data aggregation catch−weighting, the newly estimated sex−disaggregated Weight parameters are used to convert to catch in numbers (sex_disagg_comp_wt == 1).

Aleutian Islands (AI) Data Use:

This model uses data from the Aleutian Islands (AI) −− no data is being removed (drop_AI == 0).

Longline Survey Index Construction:

There was a longline survey in 2026, so there is a terminal year LLS RPN value (LLS_active == 1).

The extrapolated values for the western AI longline survey stations were NOT included in the LLS RPN index (drop_WAI_LLS == 1).

The longline survey did NOT use off−year extrapolations for the BS (even years) or AI (odd years).
Only the non−extrapolated area RPNs were included in the index (drop_LLS_extrap == 1 & do_LLS_carry == 0).
The FAA model is not using a carryover version of the LLS RPN index (do_spatial == 1 & do_carry_FAA == 0).

The SPoRC model is being passed the index SEs, no conversions needed (do_cont_SE == 1).

Composition Data Structure and ISS:

Composition proportions by age/length are aggregated across regions after weighting by catch in numbers or RPNs.

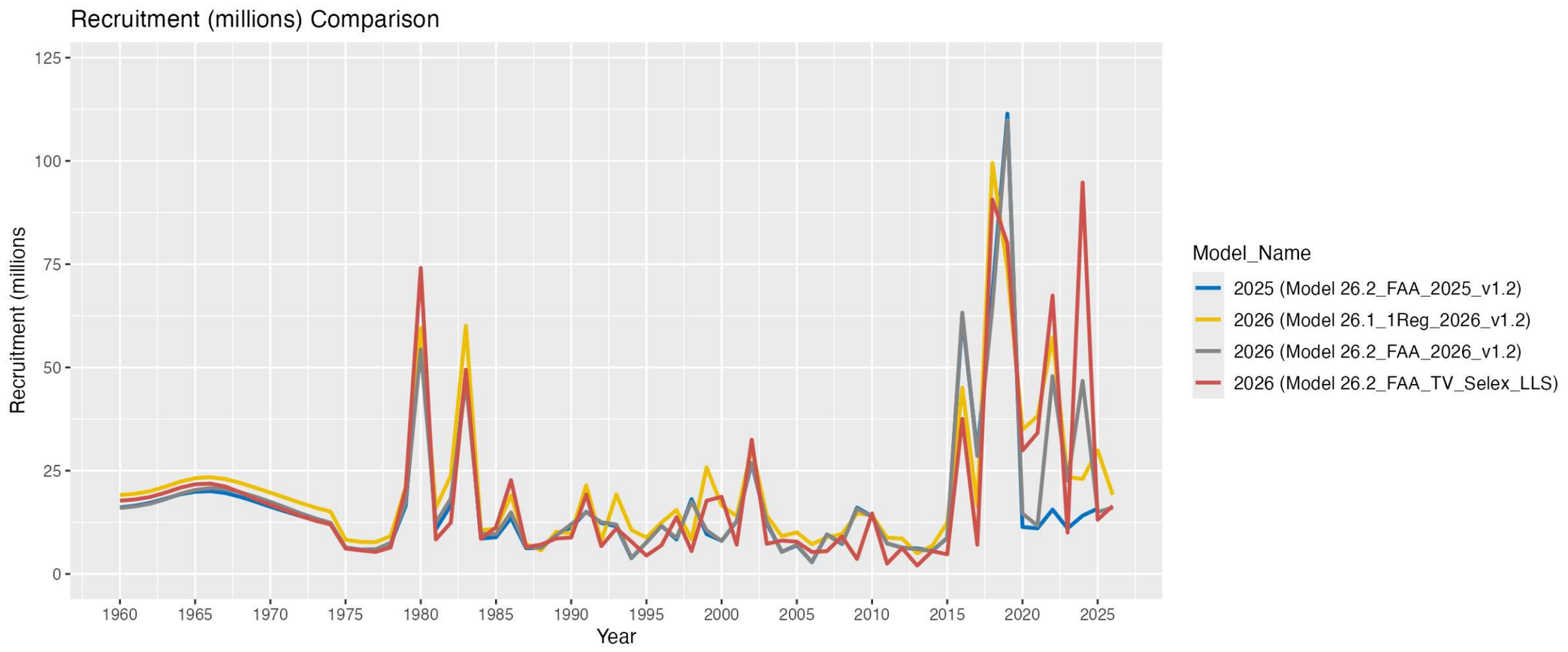
Age and length compositions are fit 'joint' by sex, but lengths are dropped for years when ages exist (JPN LLS lengths are kept for all years) (do_spatial > 0 & sex_age_comps == 5).
The Input Sample Size (ISS) for composition data is scaled to a maximum value across years/sexes/fleets to address interannual variability (do_spatial == 1 & do_scale_ISS == 1).

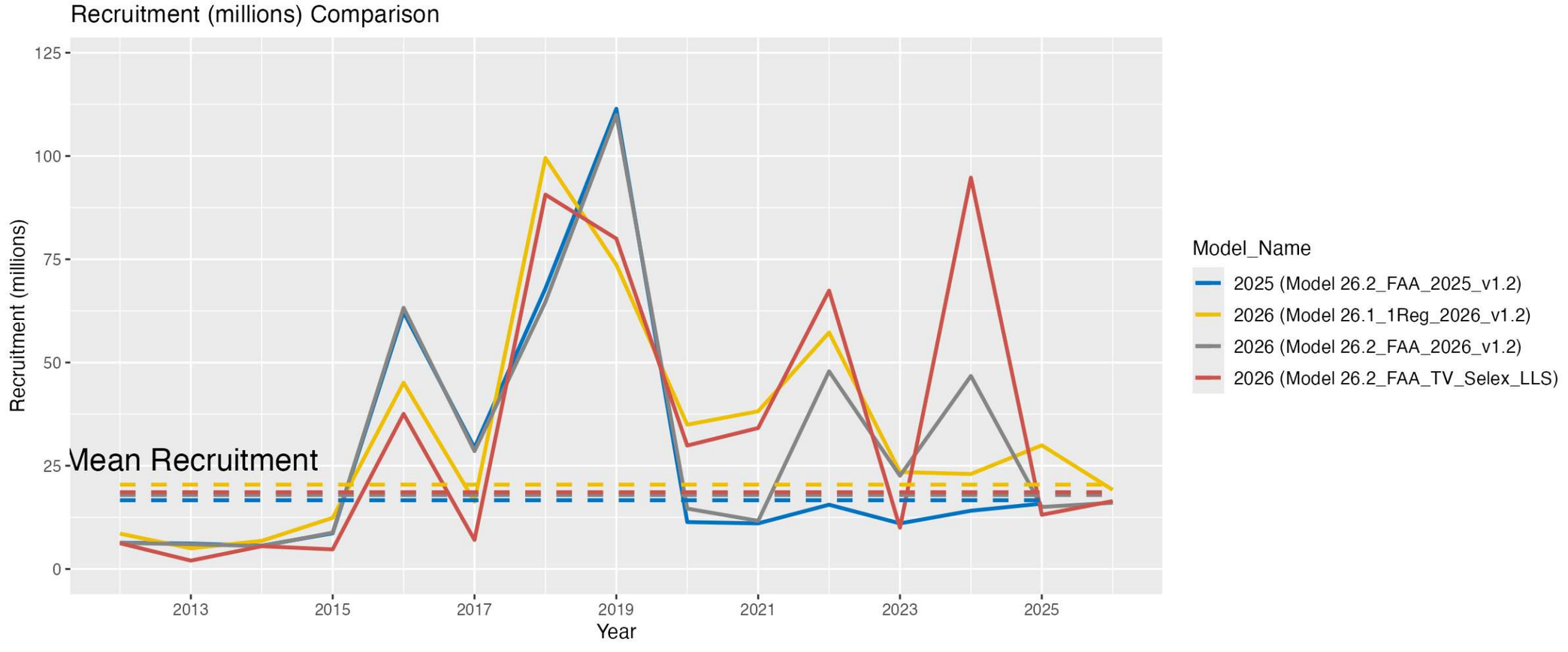
Length Composition Data Use:

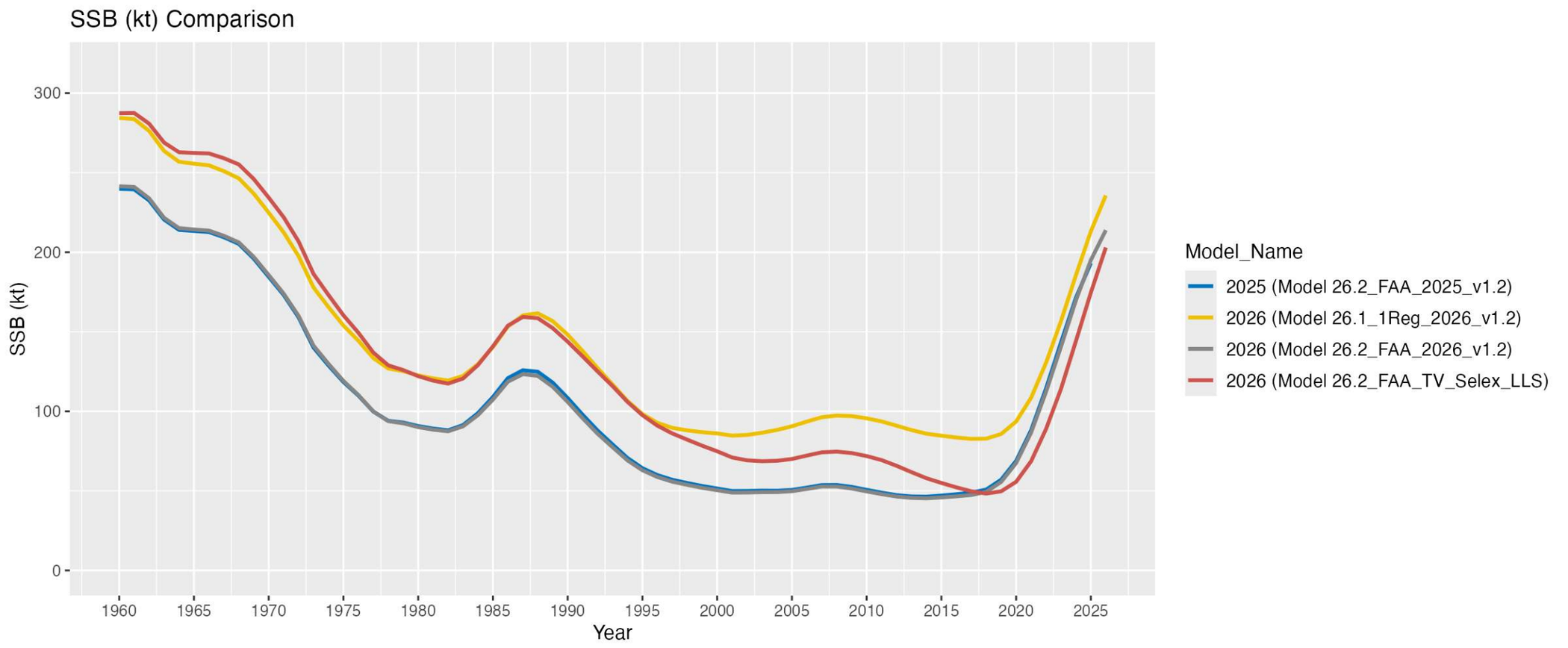
The longline survey length composition data is weighted/calculated using the raw length data and weighted by area RPNs consistent with the age composition data (LLS_length_comp_source == 1).
No longline survey length data are fit in the model (this does not apply to JPN LLS lengths) −− only age data are used (fit_LLS_len == 0).
No fixed gear fishery length data are fit in the model (this does not apply to trawl fishery lengths) −− only age data are used (fit_LLF_len == 0).

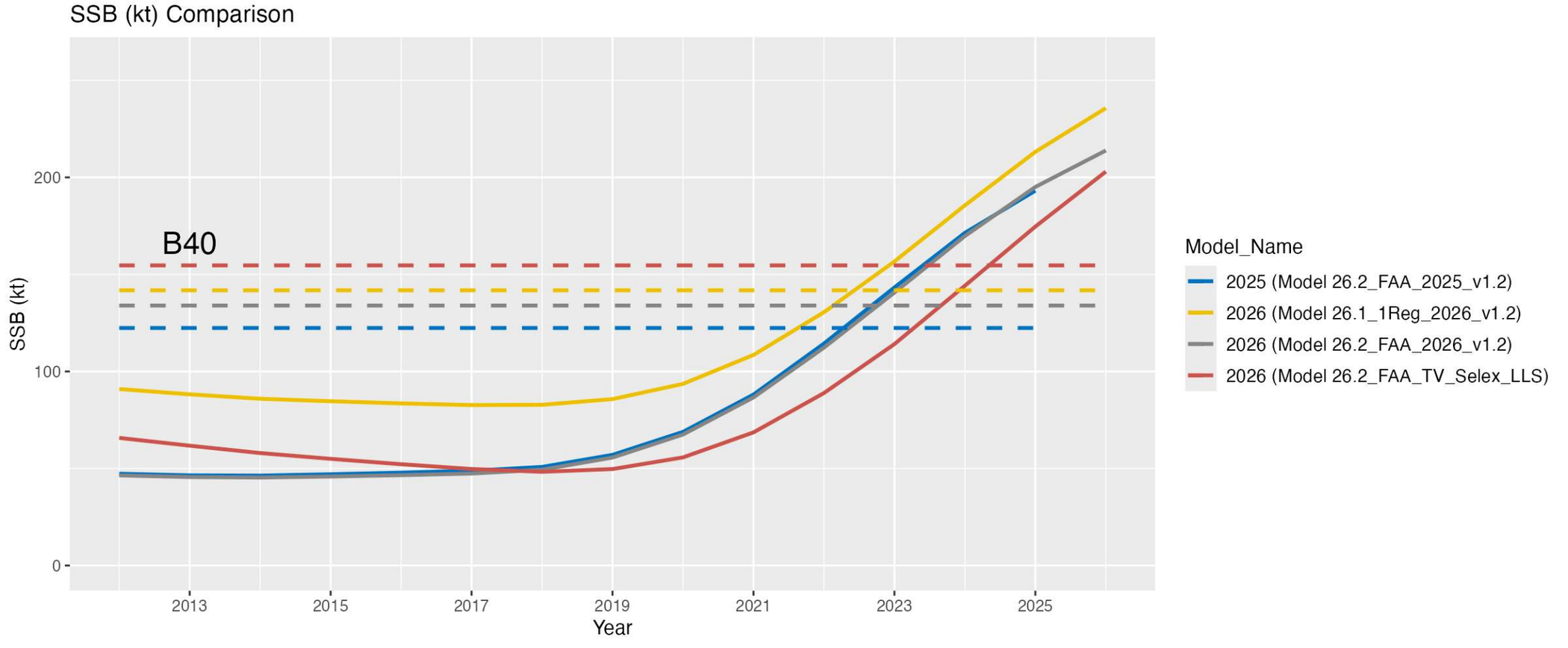
General Data Fitting Notes:

Fishery whale depredation estimates were fixed/static at the previous year's values (whale_dep_static == 1).
The GOA trawl survey data was NOT fit in this model (use_trwl_srvy == 0).
The fishery CPUE index was NOT fit in this model (use_CPUE == 0).
Non−fishery catch (e.g., scientific removals) WAS included in the model as fixed gear catch, where only LLS removals were available for terminal year (add_non_fish_catch == 1 & non_fsh_catch_updated == 0).

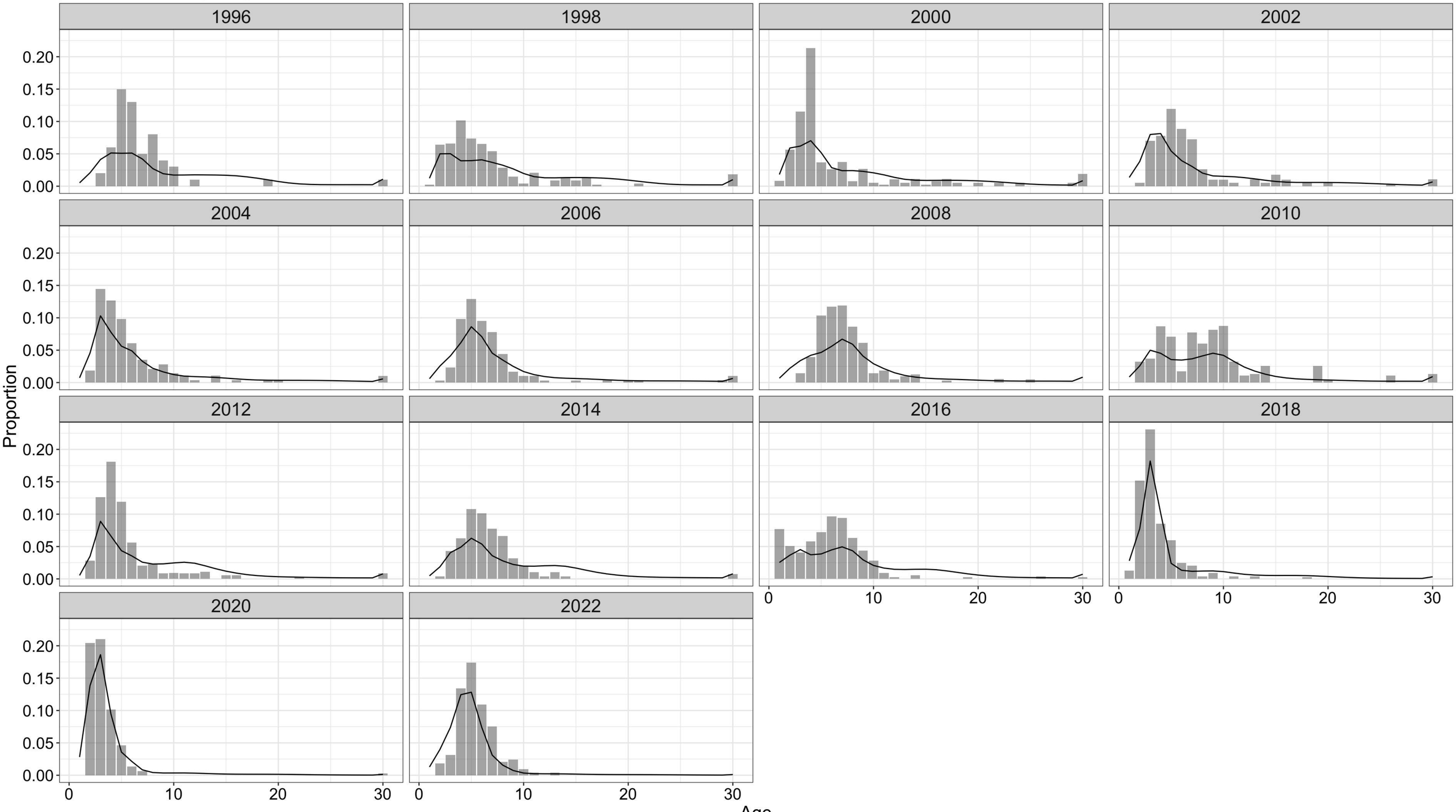

Recruitment (millions) Comparison
Recruitment (millions)
125
100
75
50
25
0
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
2025
Year
Model_Name
2025 (Model 26.2_FAA_2025_v1.2)
2026 (Model 26.1_1Reg_2026_v1.2)
2026 (Model 26.2_FAA_2026_v1.2)
2026 (Model 26.2_FAA_TV_Selex_LLS)


Recruitment (millions) Comparison
Recruitment (millions)
125
100
75
50
25
0
Mean Recruitment
2013
2015
2017
2019
2021
2023
2025
Year
Model_Name
2025 (Model 26.2_FAA_2025_v1.2)
2026 (Model 26.1_1Reg_2026_v1.2)
2026 (Model 26.2_FAA_2026_v1.2)
2026 (Model 26.2_FAA_TV_Selex_LLS)


SSB (kt) Comparison
300
200
100
0
SSB (kt)
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
2025
Year
Model_Name
2025 (Model 26.2_FAA_2025_v1.2)
2026 (Model 26.1_1Reg_2026_v1.2)
2026 (Model 26.2_FAA_2026_v1.2)
2026 (Model 26.2_FAA_TV_Selex_LLS)


SSB (kt) Comparison
200
100
0
SSB (kt)
B40
2013
2015
2017
2019
2021
2023
2025
Year
Model_Name
2025 (Model 26.2_FAA_2025_v1.2)
2026 (Model 26.1_1Reg_2026_v1.2)
2026 (Model 26.2_FAA_2026_v1.2)
2026 (Model 26.2_FAA_TV_Selex_LLS)

AI LL Survey Age Comps Female AK

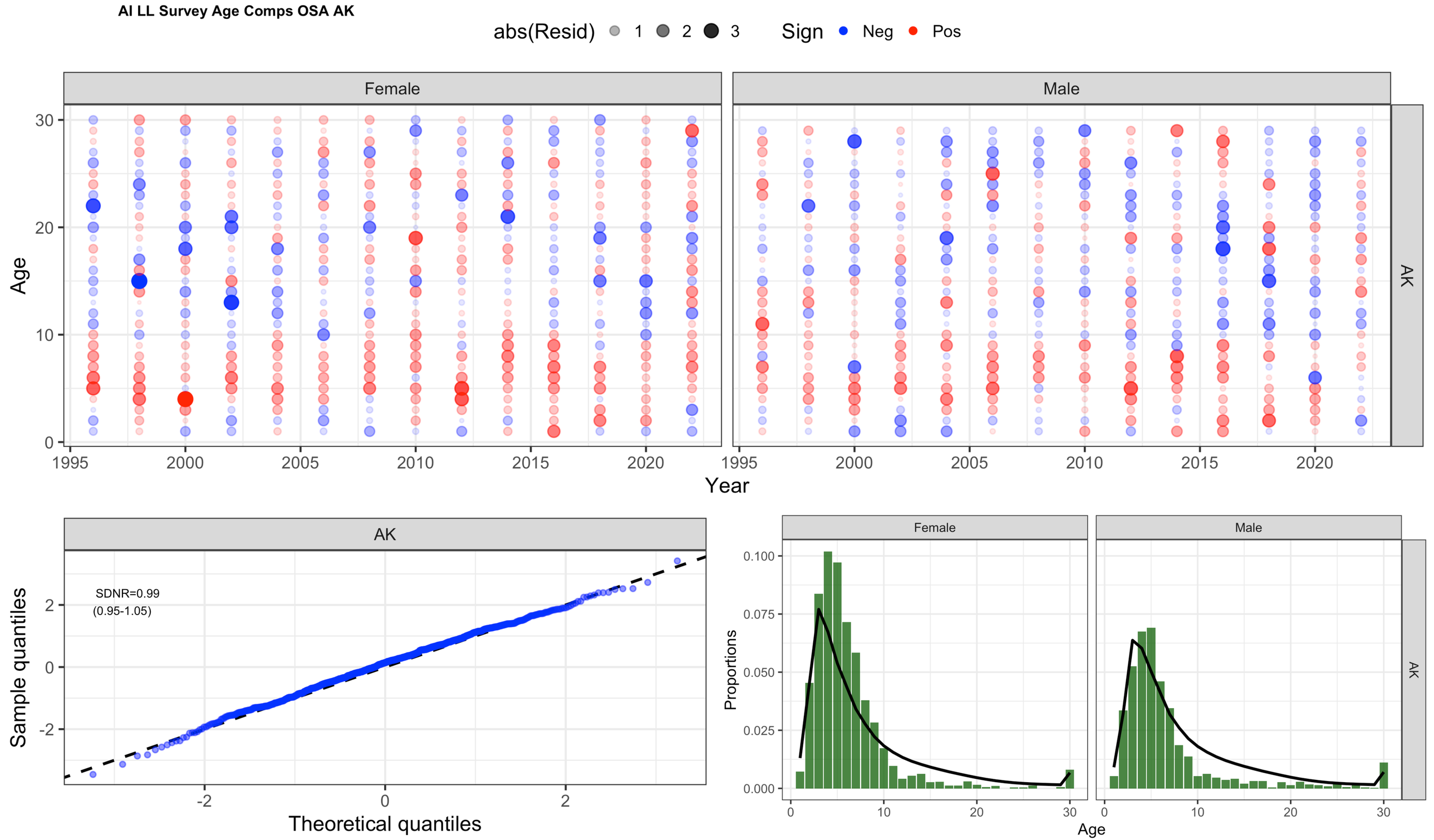


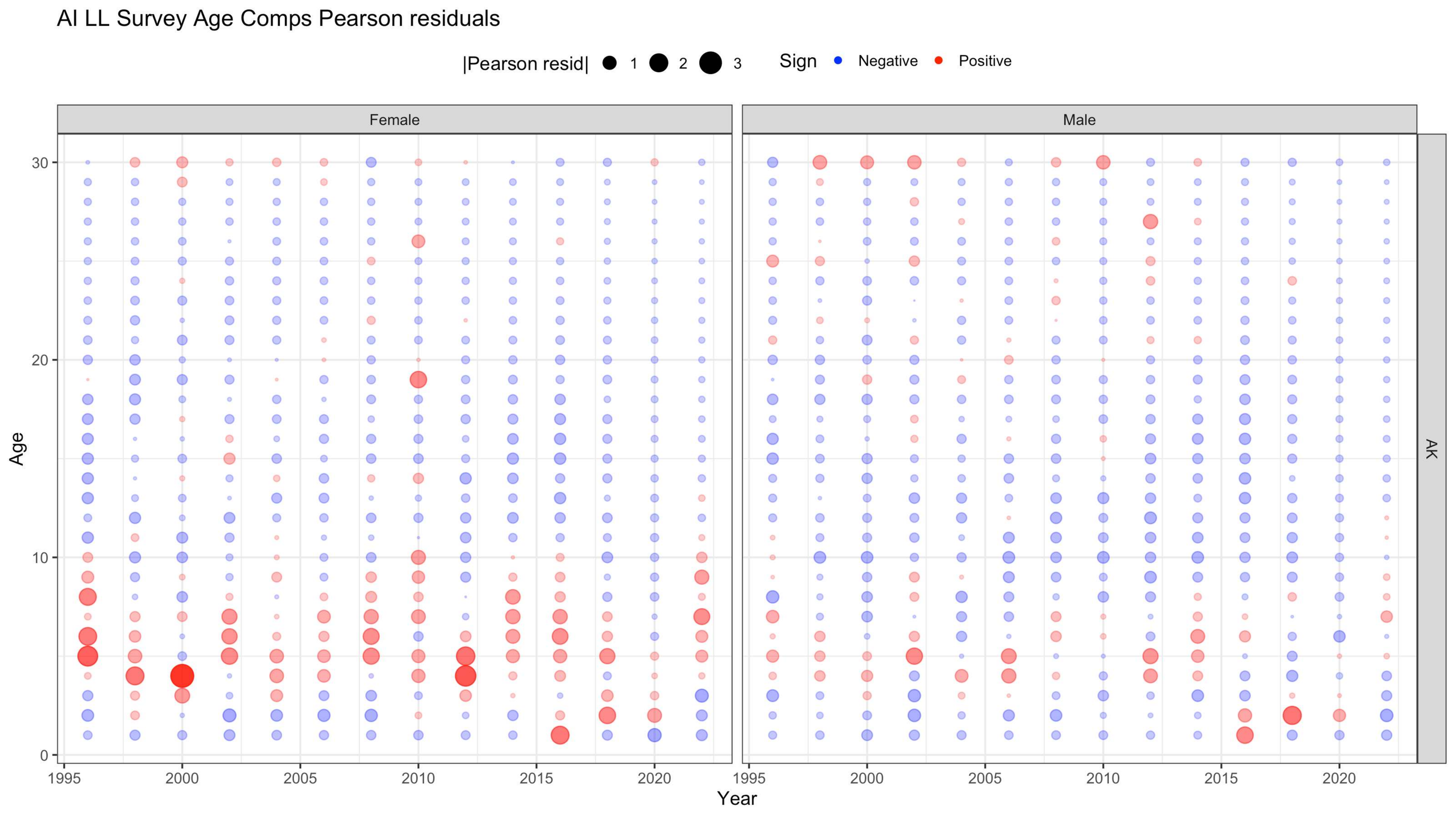
AI LL Survey Age Comps Male AK

1996 1998 2000 2002
2004 2006 2008 2010
2012 2014 2016 2018
2020 2022

Proportion
0.15 0.10 0.05 0.00
0 10 20 30
Age

AI LL Survey Age Comps OSA AK




AI LL Survey Age Comps Pearson residuals
|Pearson resid| 1 2 3
Sign Negative Positive
Female
Male
AK
Age
Year
0
10
20
30
1995
2000
2005
2010
2015
2020

Model 26.2_FAA_TV_Selex_LLS Population Population 1

Female

Male

AK

Spawning Weight-at-Age (kg)

0.00

0.02

0.04

0.06

0.08

0

10

20

30

Age

Model 26.2_FAA_TV_Selex_LLS Population Population 1

Female

Male

AK

Maturity at Age

1.00

0.75

0.50

0.25

0.00

0

10

20

30

Age

Model 26.2_FAA_TV_Selex_LLS Population Population 1

Female

Male

AK

Spawning Weight at Age

6

4

2

0

0

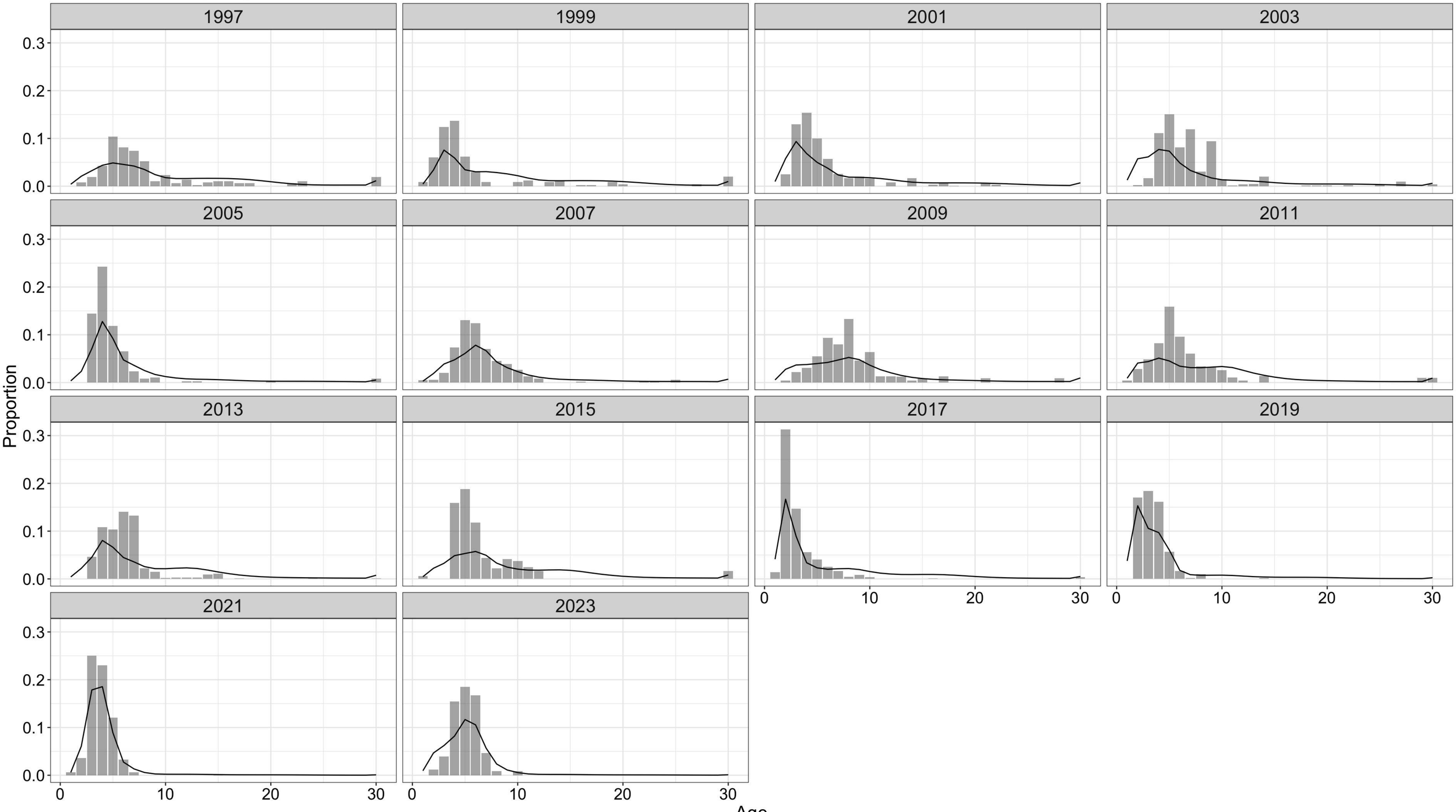
10

20

30

Age

BS LL Survey Age Comps Female AK



BS LL Survey Age Comps Male AK

1997 1999 2001 2003
2005 2007 2009 2011
2013 2015 2017 2019
2021 2023

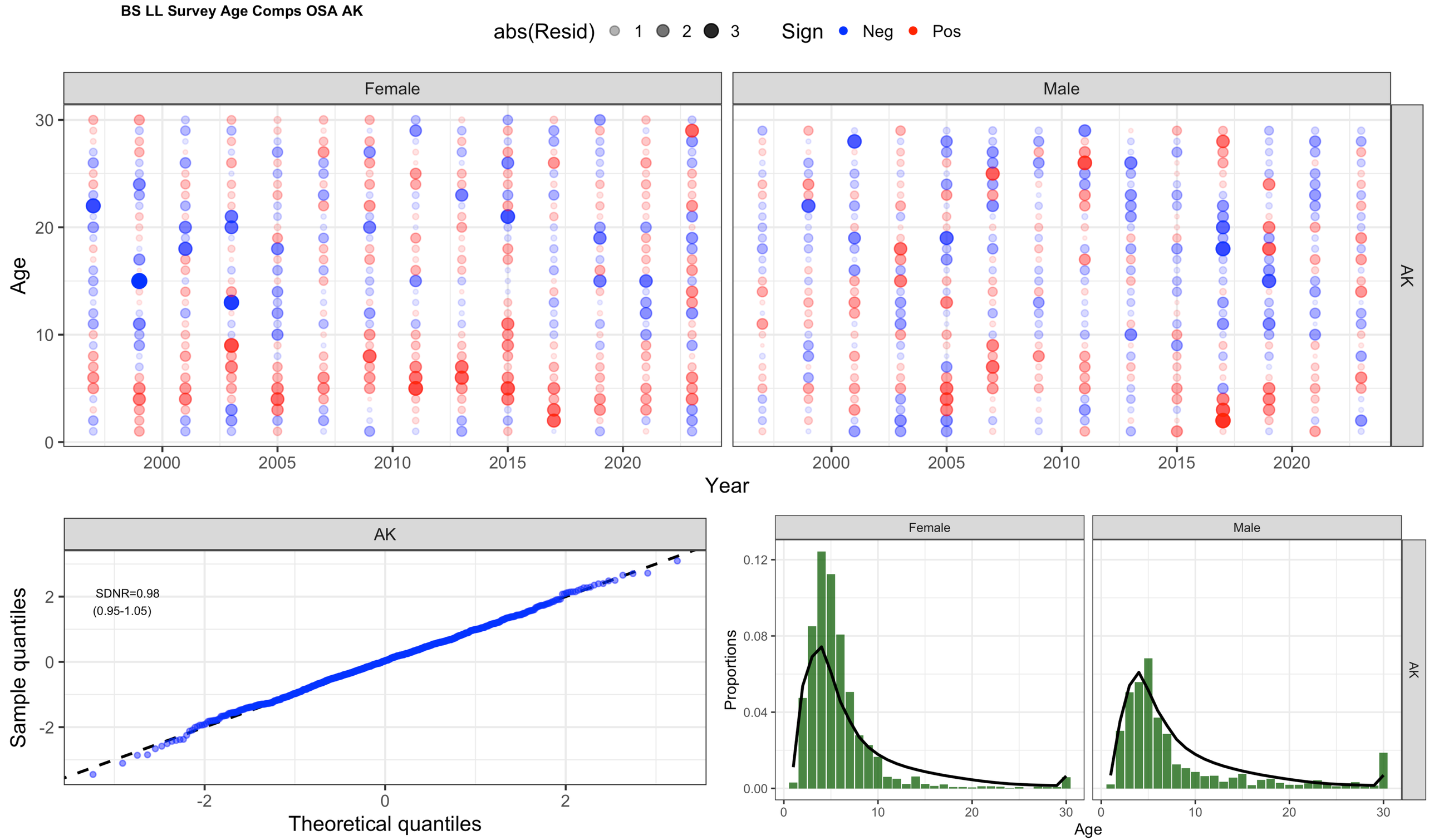
Proportion

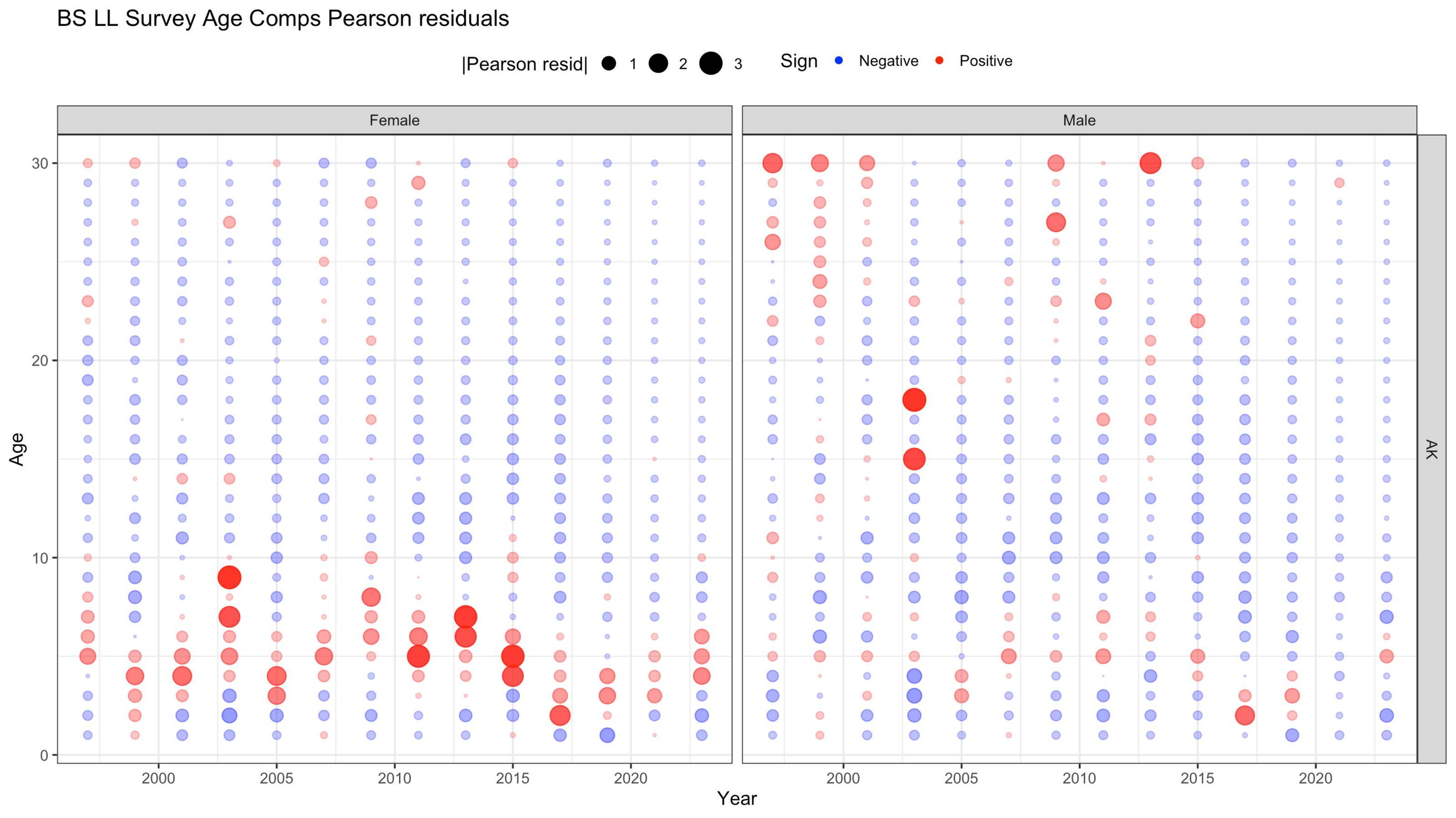
0.00 0.05 0.10 0.15

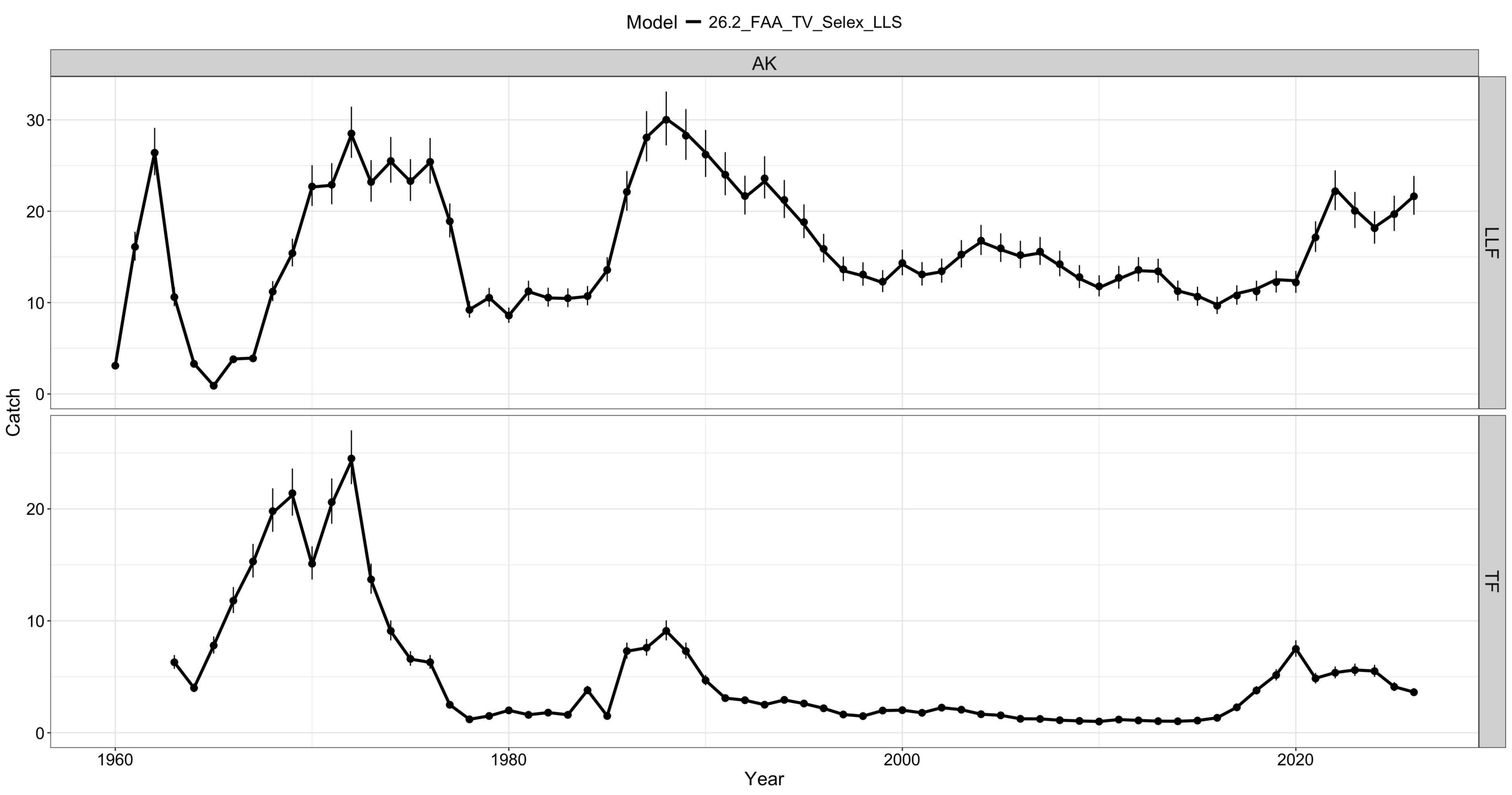
0 10 20 30

Age

BS LL Survey Age Comps OSA AK




BS LL Survey Age Comps Pearson residuals
|Pearson resid| 1 2 3
Sign Negative Positive
Female
Male
AK
Age
Year
0
10
20
30
2000
2005
2010
2015
2020


Model 26.2_FAA_TV_Selex_LLS
AK
LLF
TF
Catch
Year
0
10
20
30
1960
1980
2000
2020

AK

26.2_FAA_TV_Selex_LLS

JPN LLS Lengths
JPN LLS Index
GOA LLS Index
AI LLS Index
BS LLS Index
JPN LLS Ages
GOA LLS Ages
AI LLS Ages
BS LLS Ages
TF Lengths
TF Catch
LLF Catch
LLF Ages

1960
1980
2000
2020

Year

GOA LL Survey Age Comps Female AK

1996 1997 1998 1999 2000 2001
2002 2003 2004 2005 2006 2007
2008 2009 2010 2011 2012 2013
2014 2015 2016 2017 2018 2019
2020 2021 2022 2023 2025

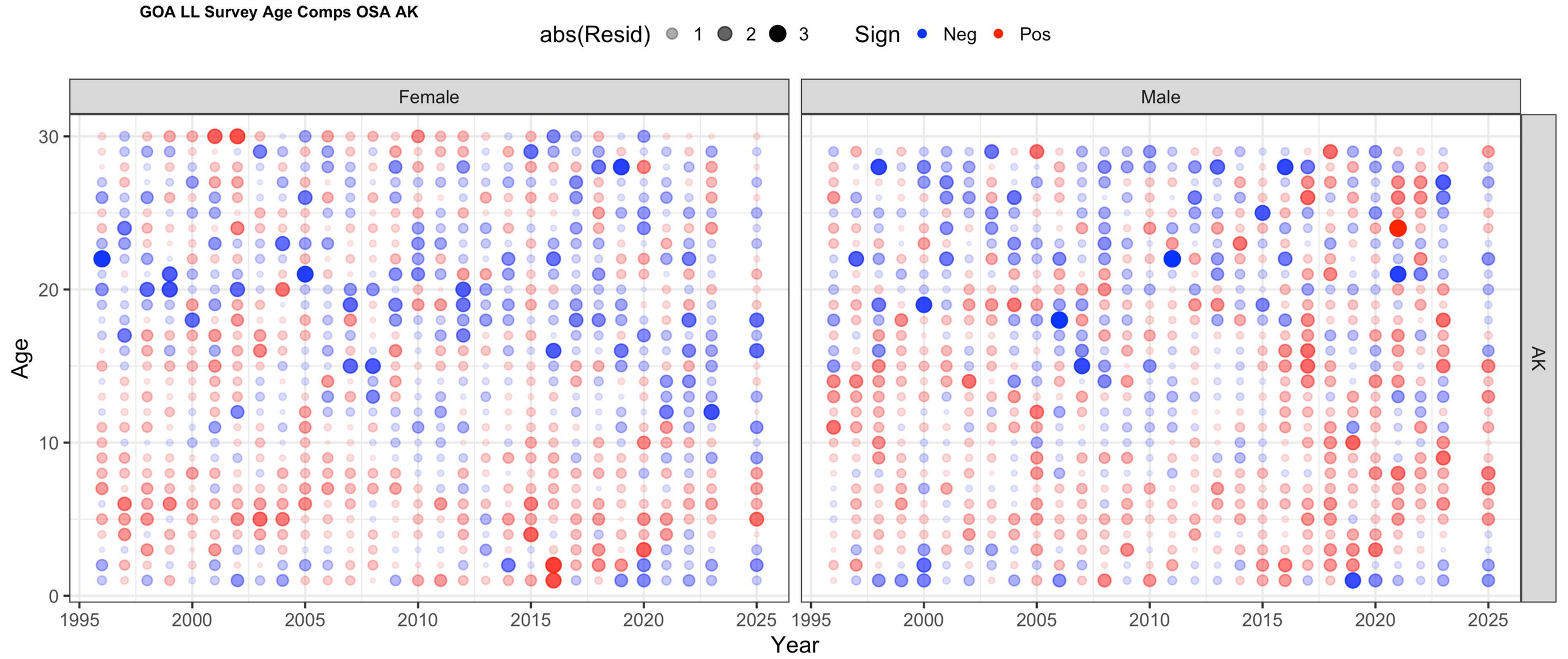
Proportion
0.0 0.1 0.2
0 10 20 30
Age

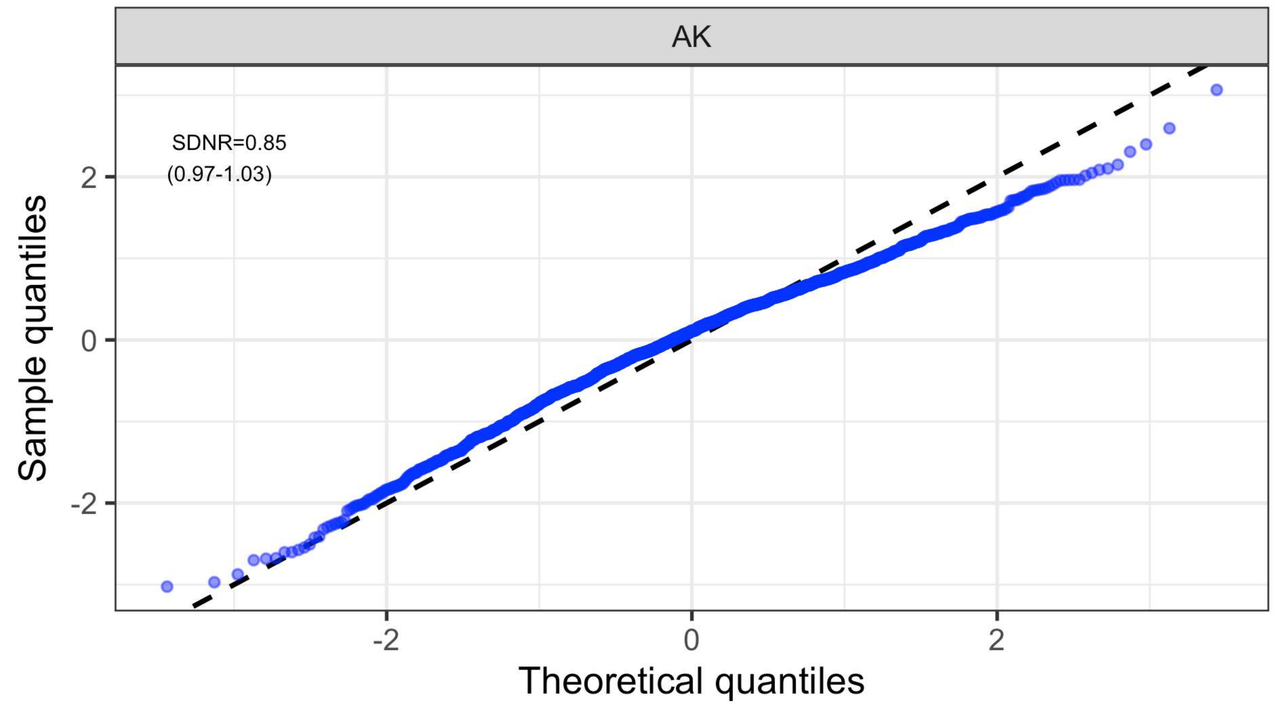
GOA LL Survey Age Comps Male AK

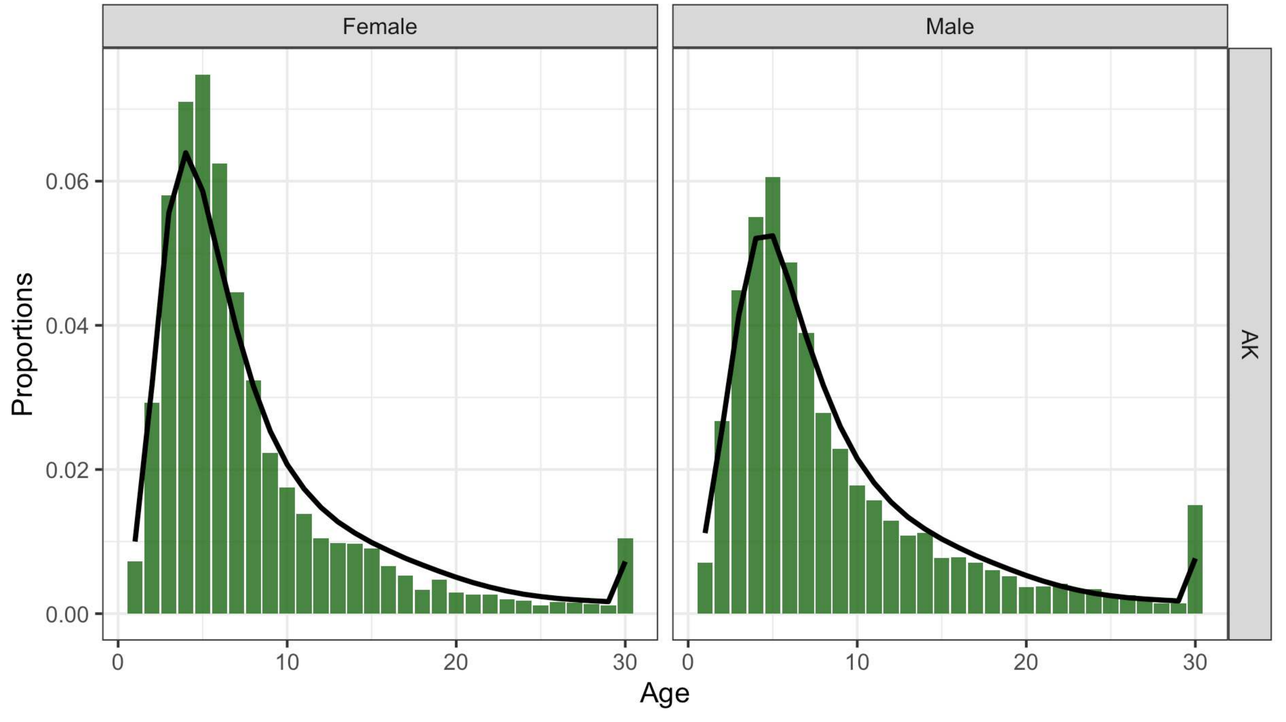
1996 1997 1998 1999 2000 2001
2002 2003 2004 2005 2006 2007
2008 2009 2010 2011 2012 2013
2014 2015 2016 2017 2018 2019
2020 2021 2022 2023 2025

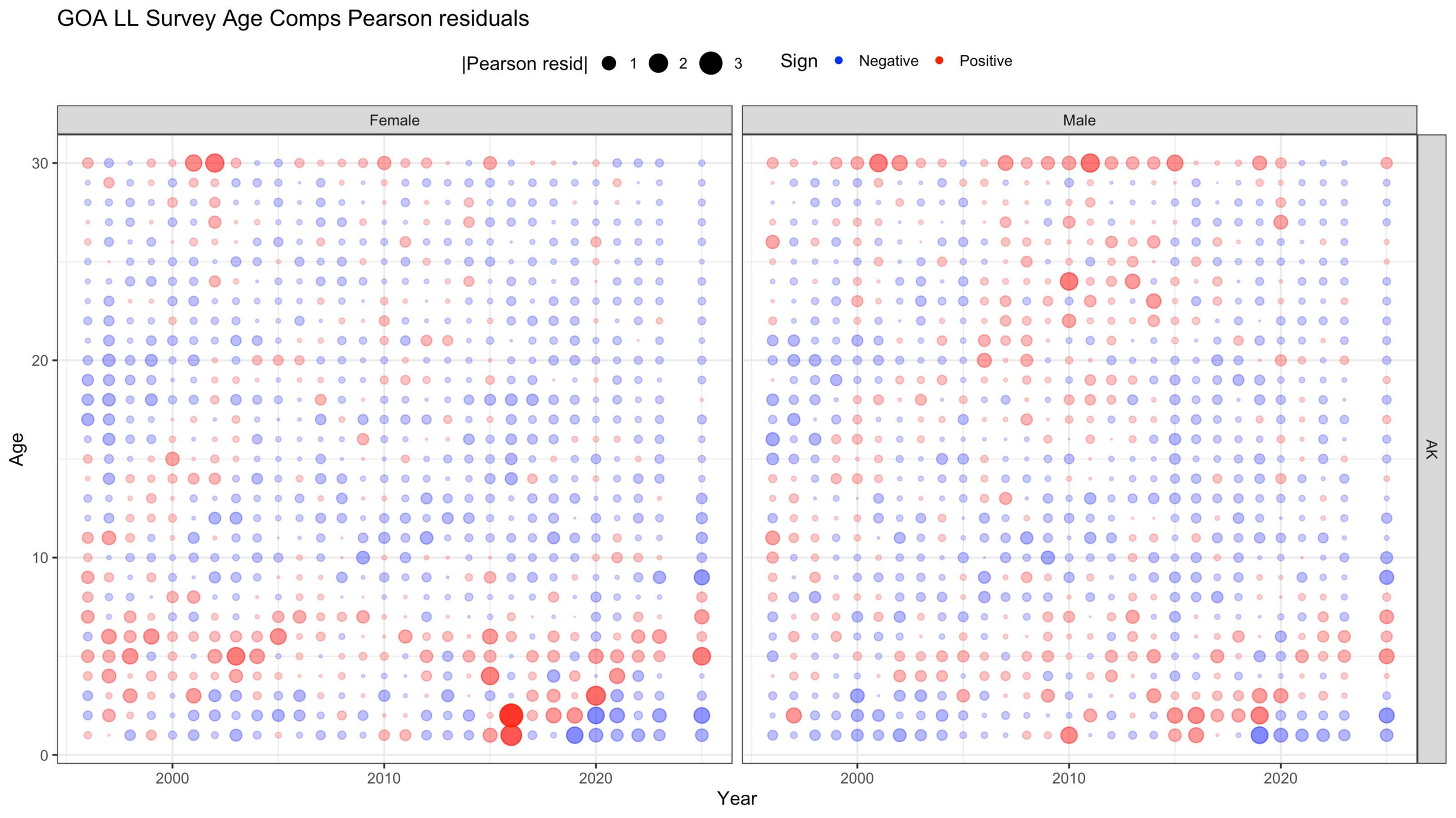
Proportion
0.00 0.05 0.10 0.15
0 10 20 30
Age

GOA LL Survey Age Comps OSA AK








GOA LL Survey Age Comps Pearson residuals
|Pearson resid| 1 2 3
Sign Negative Positive
Female
Male
AK
Age
Year
0
10
20
30
2000
2010
2020

Model — 26.2_FAA_TV_Selex_LLS

BS LLS

AI LLS

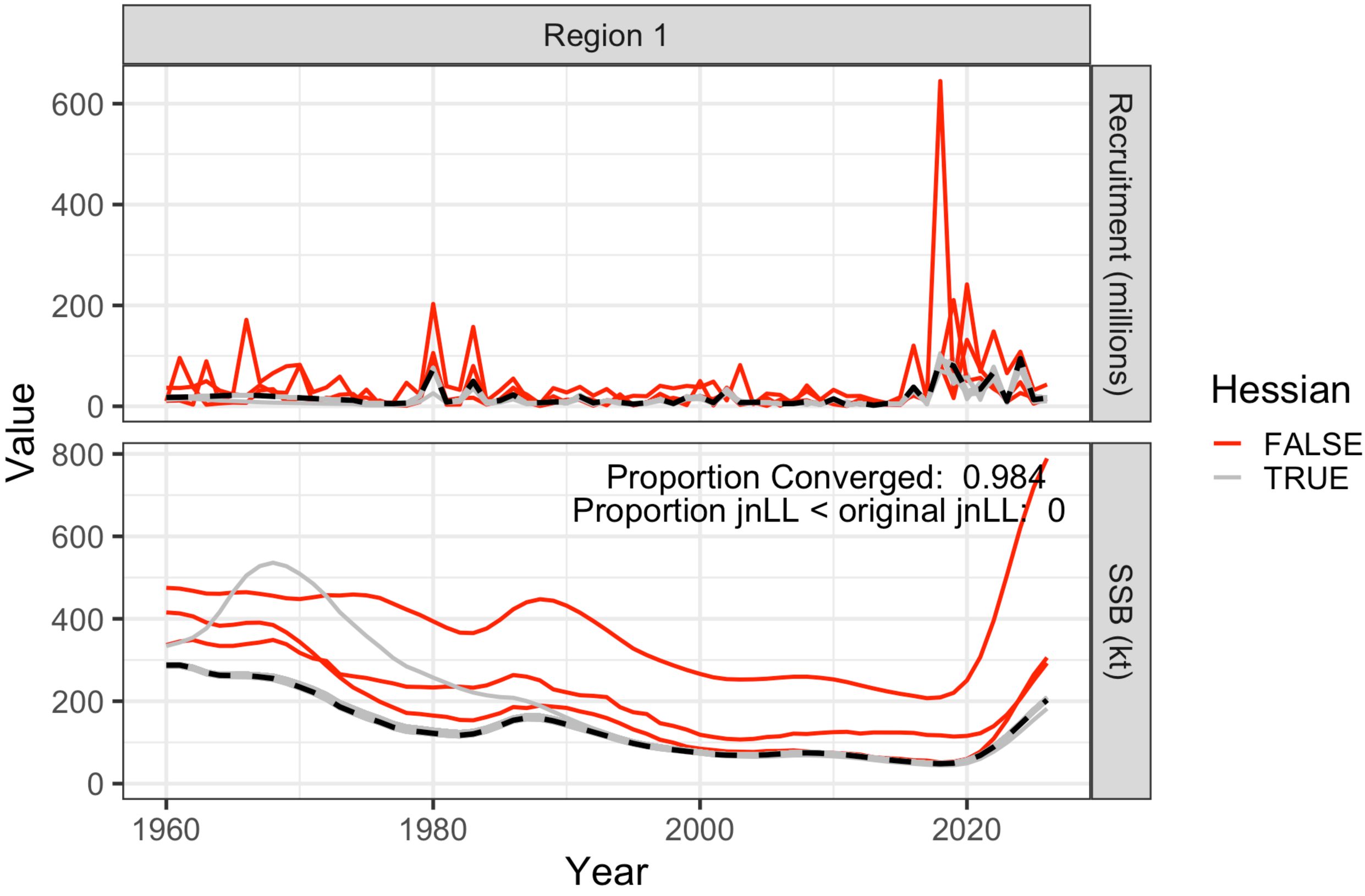
GOA LLS

JPN LLS

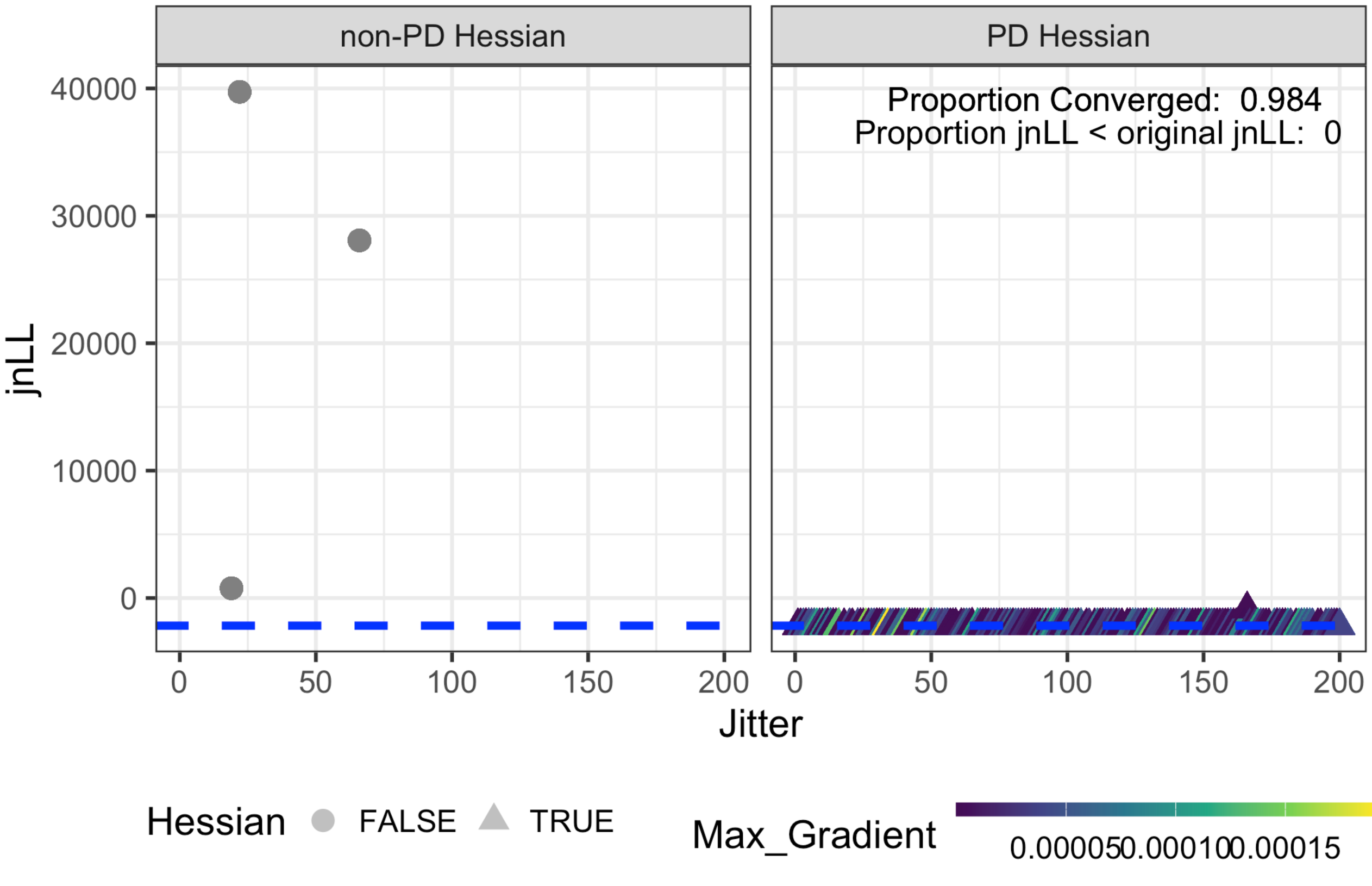
RPNs (1000s)

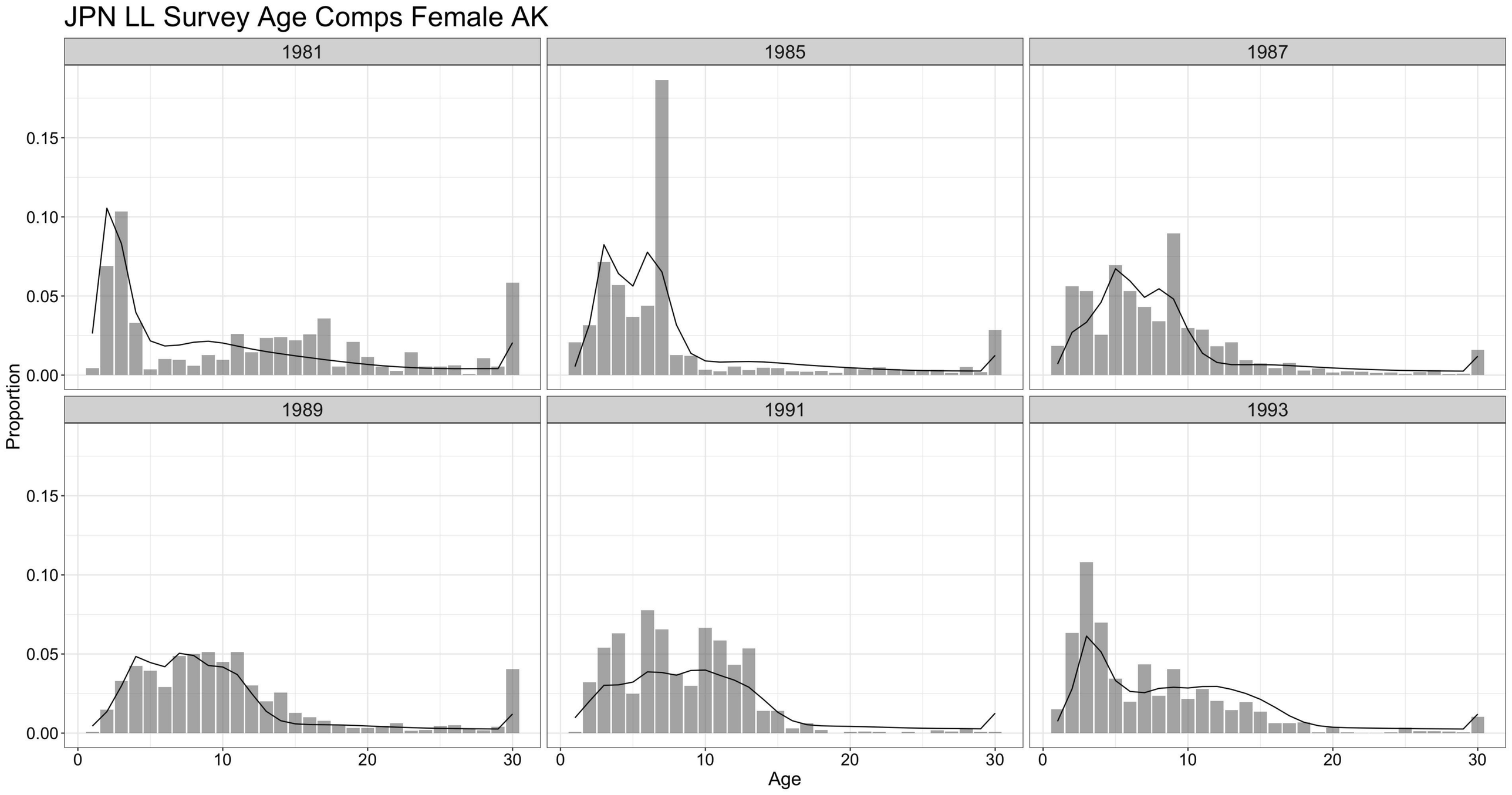
Year

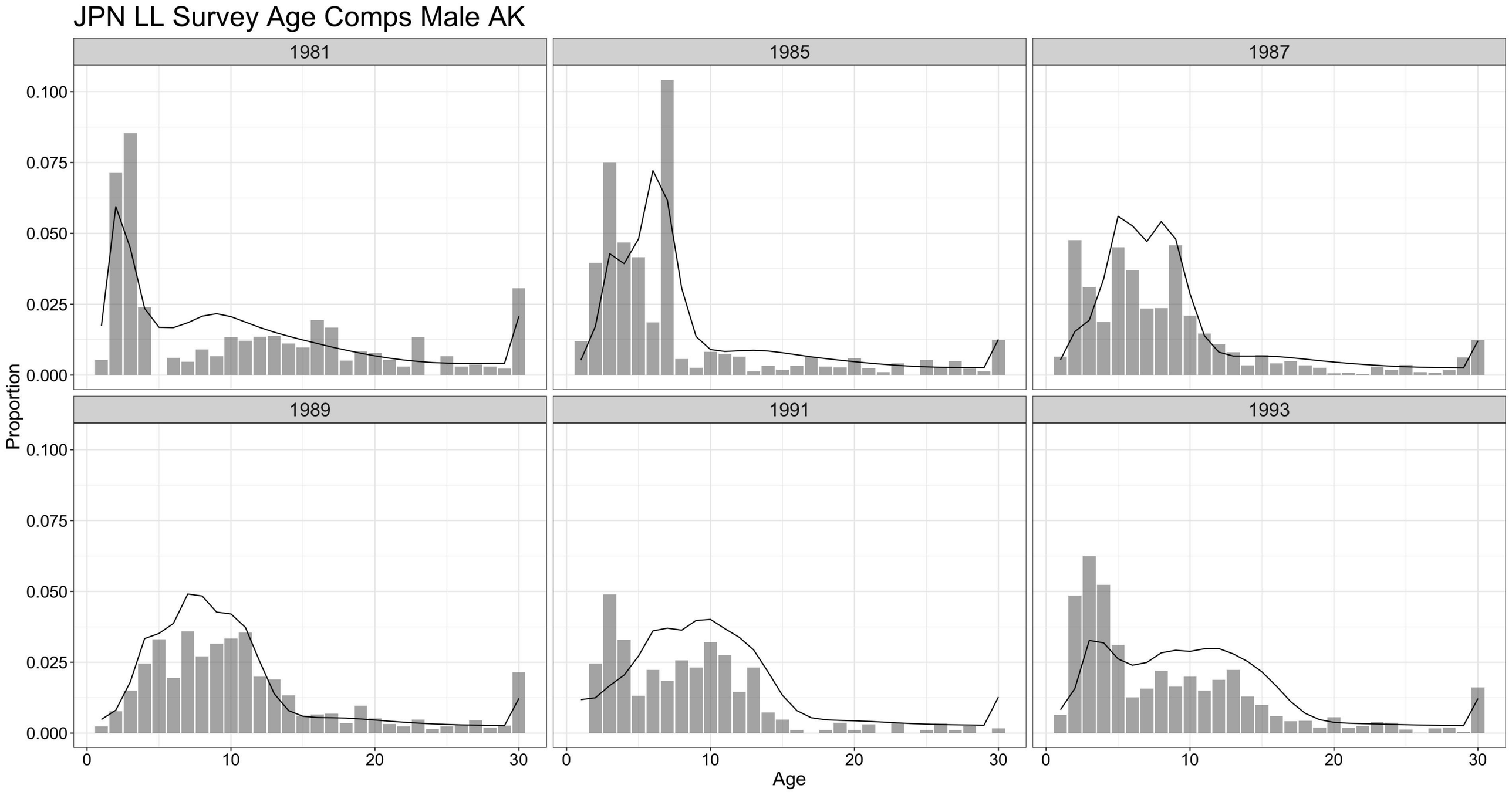
Jitter SSB and Recruitment



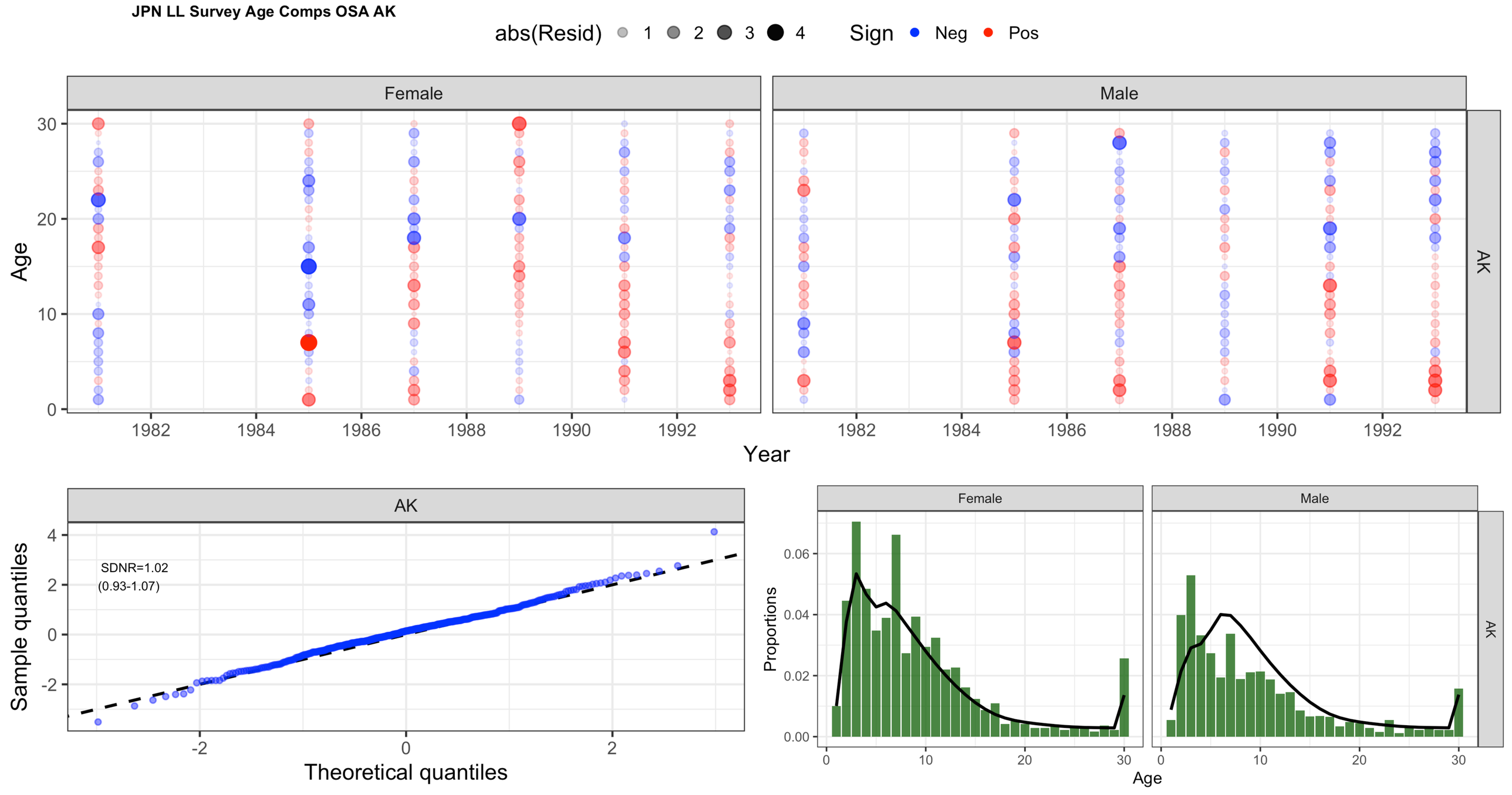
Jitter nLL and Convergence

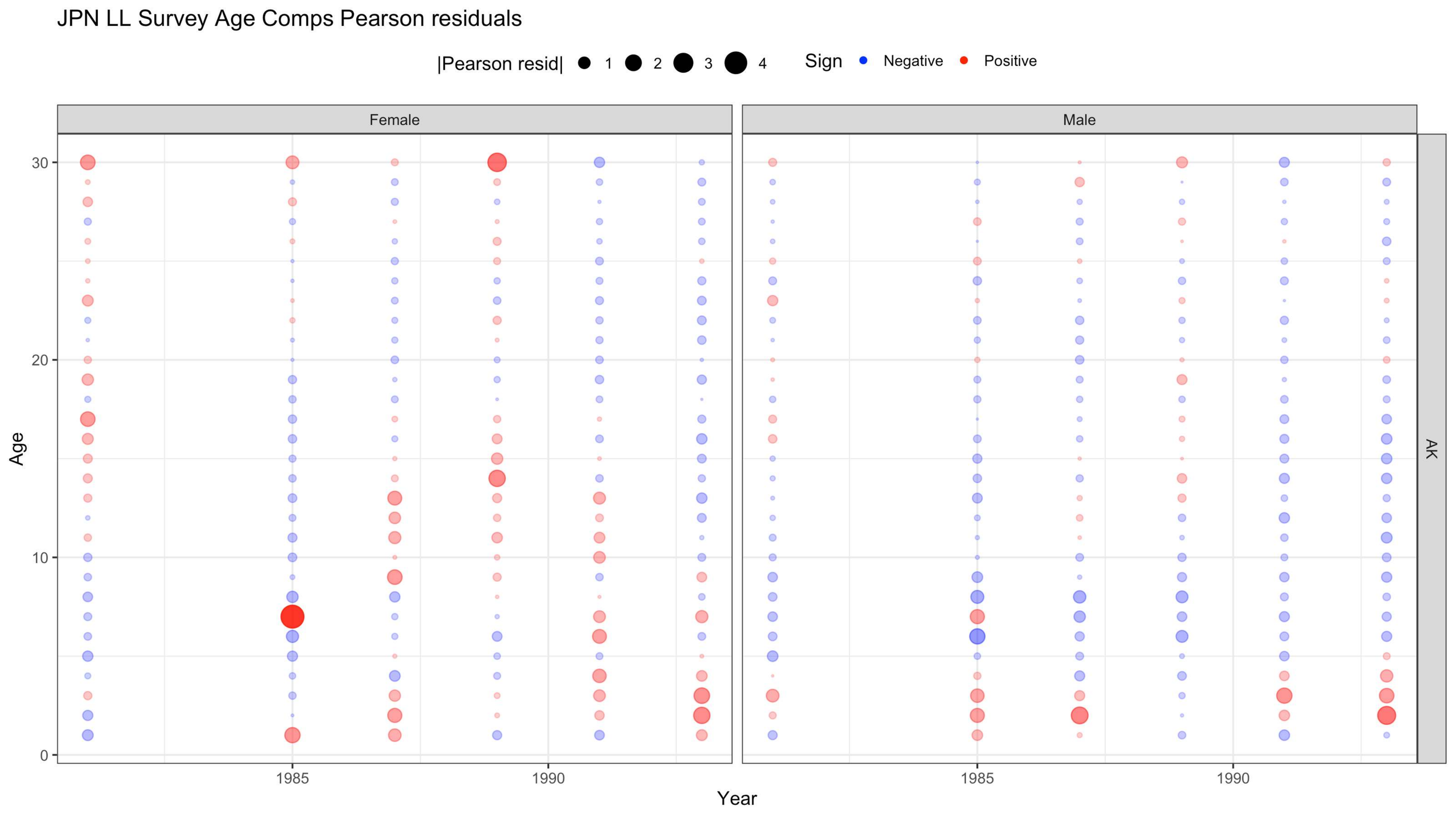



JPN LL Survey Age Comps Female AK
1981
1985
1987
1989
1991
1993
Proportion
0.00
0.05
0.10
0.15
0
10
20
30
Age


JPN LL Survey Age Comps Male AK
1981
1985
1987
1989
1991
1993
0.100
0.075
0.050
0.025
0.000
Proportion
0
10
20
30
Age

JPN LL Survey Age Comps OSA AK




JPN LL Survey Age Comps Pearson residuals
|Pearson resid| 1 2 3 4
Sign Negative Positive
Female
Male
AK
Age
Year
0
10
20
30
1985
1990
1985
1990

JPN LL Survey Length Comps Female AK

1979 1980 1981 1982
1983 1984 1985 1986
1987 1988 1989 1990
1991 1992 1993 1994

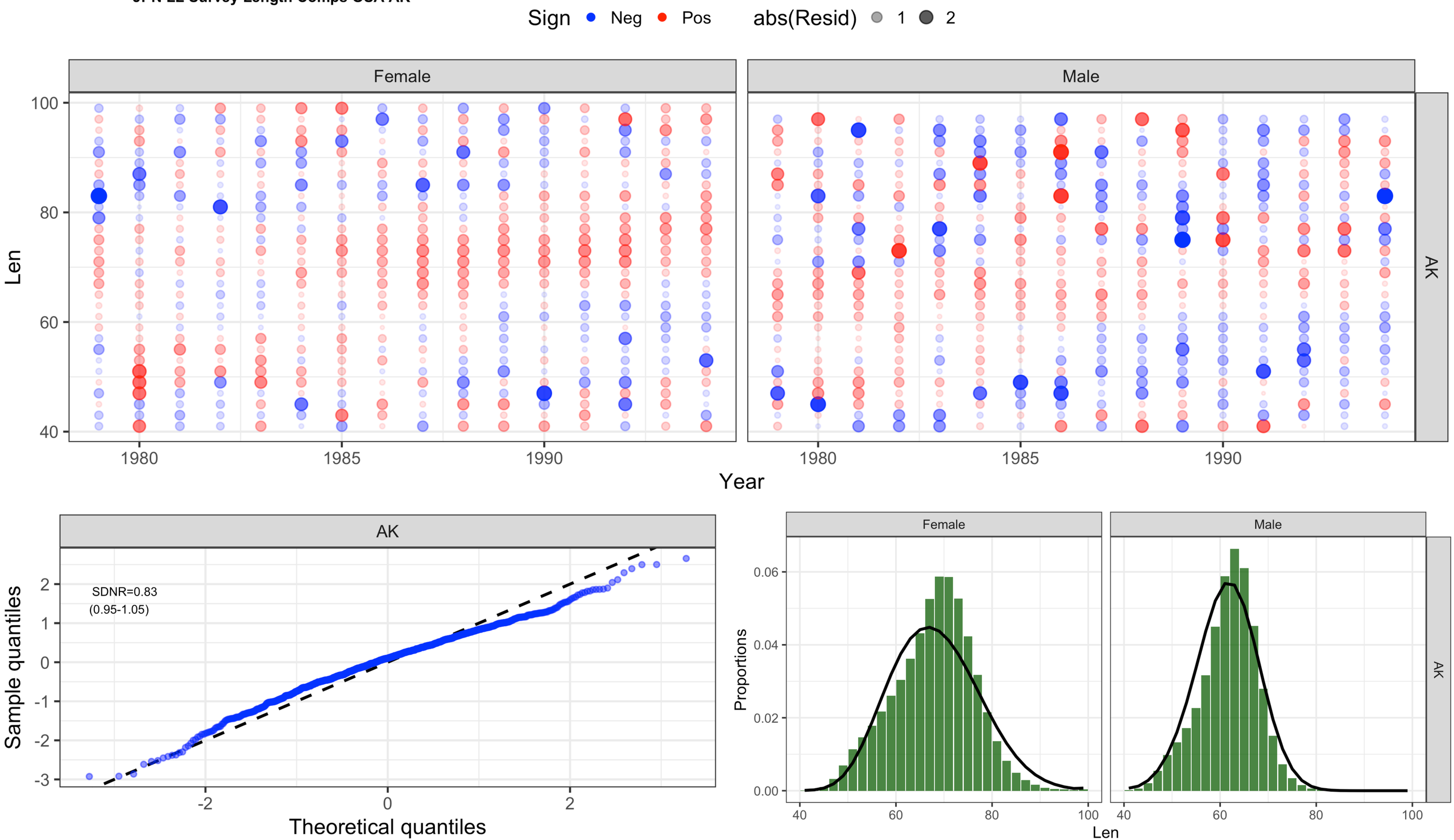
Proportion
0.00 0.02 0.04 0.06 0.08
40 60 80 100
Len

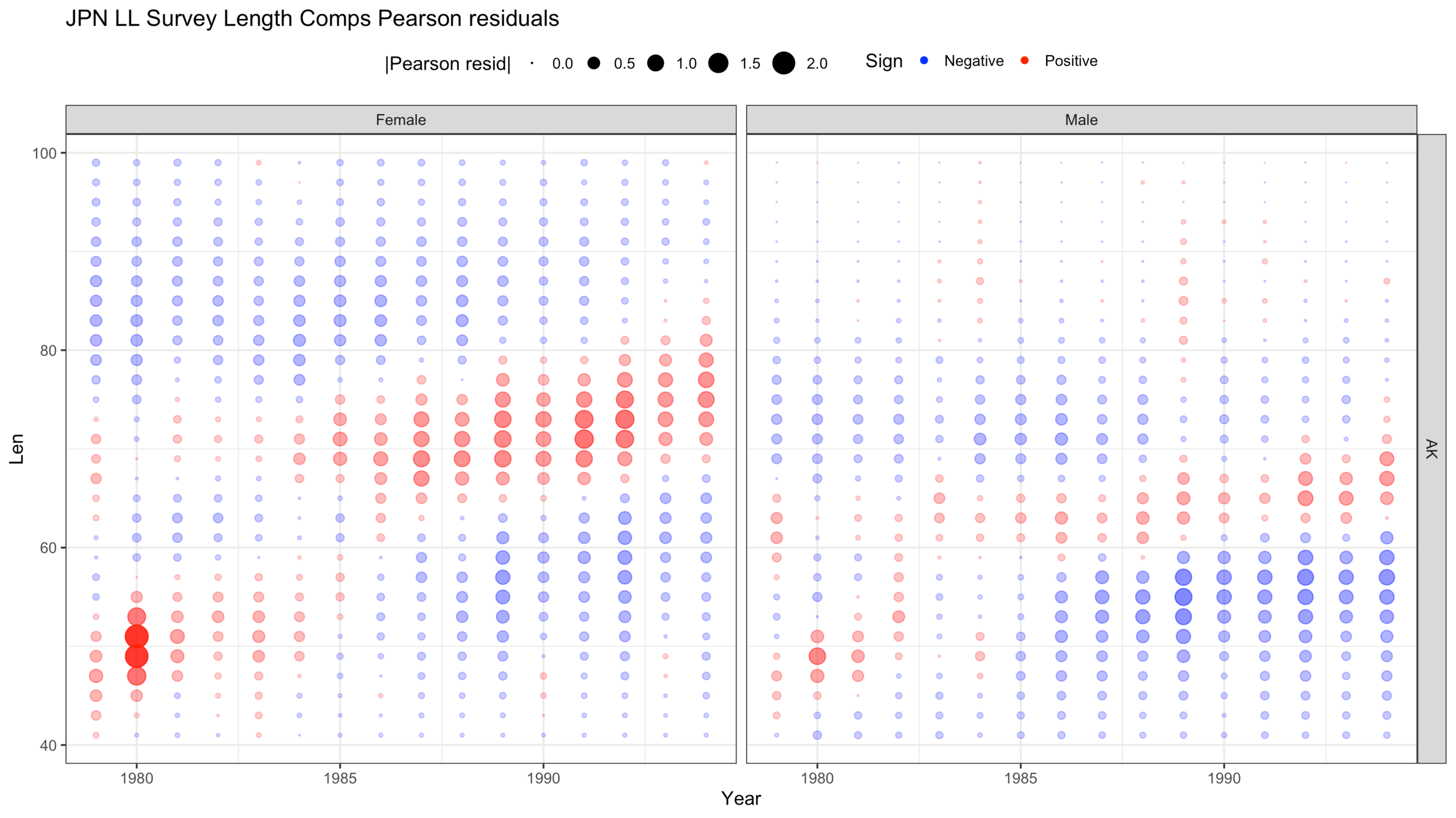
JPN LL Survey Length Comps Male AK

1979 1980 1981 1982
1983 1984 1985 1986
1987 1988 1989 1990
1991 1992 1993 1994

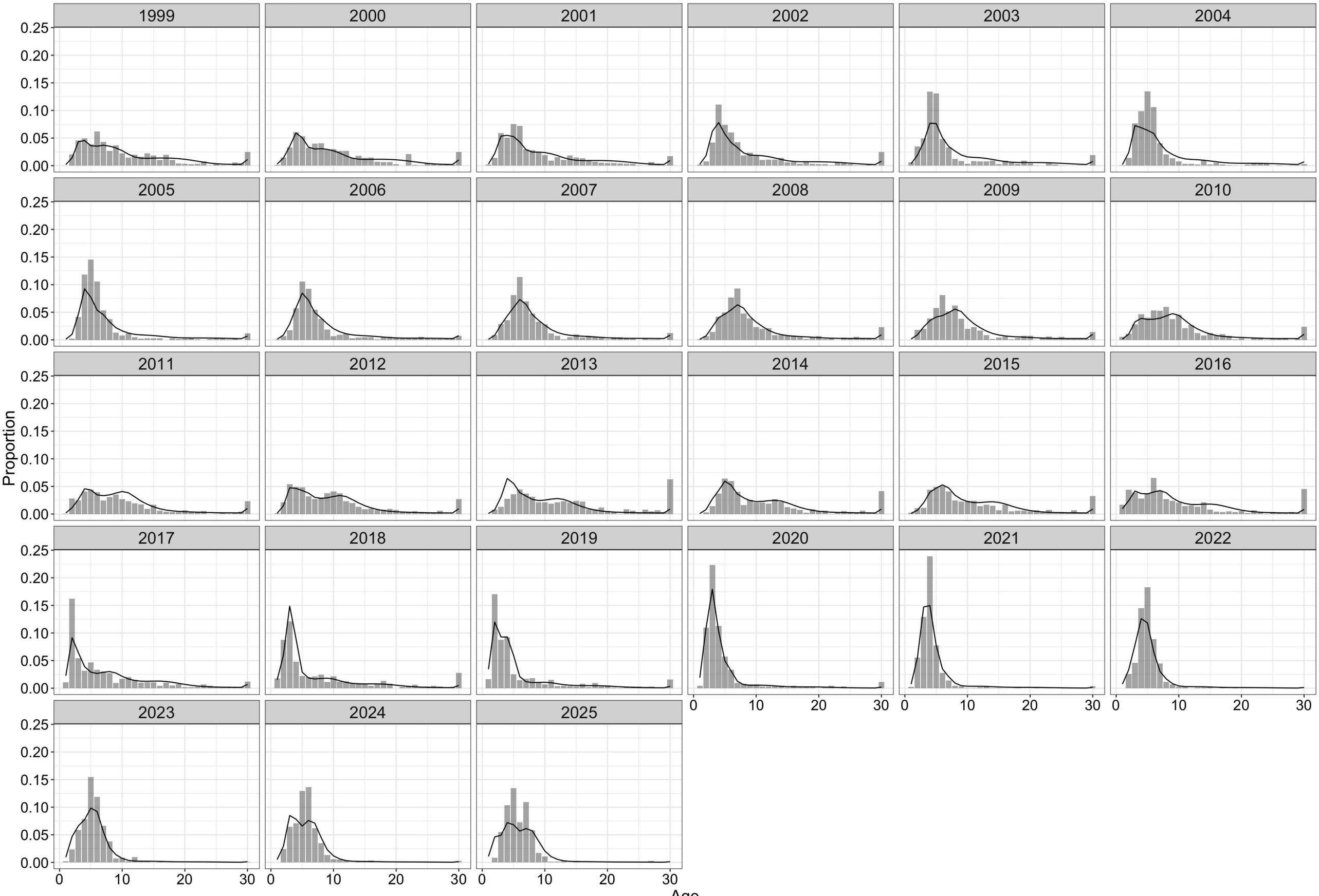
Proportion
0.00 0.02 0.04 0.06 0.08
40 60 80 100
Len

JPN LL Survey Length Comps OSA AK

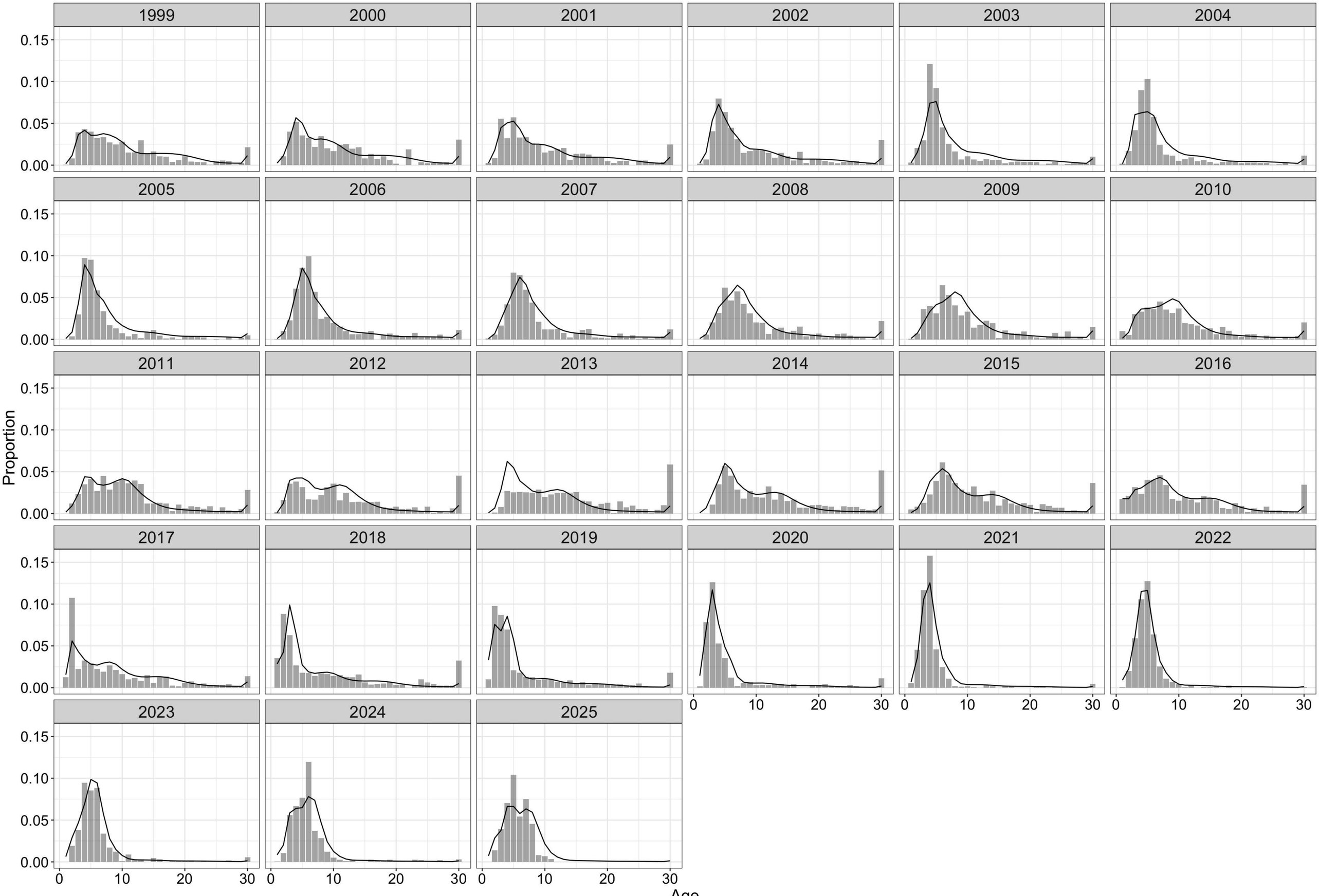



JPN LL Survey Length Comps Pearson residuals
|Pearson resid|
0.0
0.5
1.0
1.5
2.0
Sign
Negative
Positive
Female
Male
AK
Len
100
80
60
40
1980
1985
1990
Year

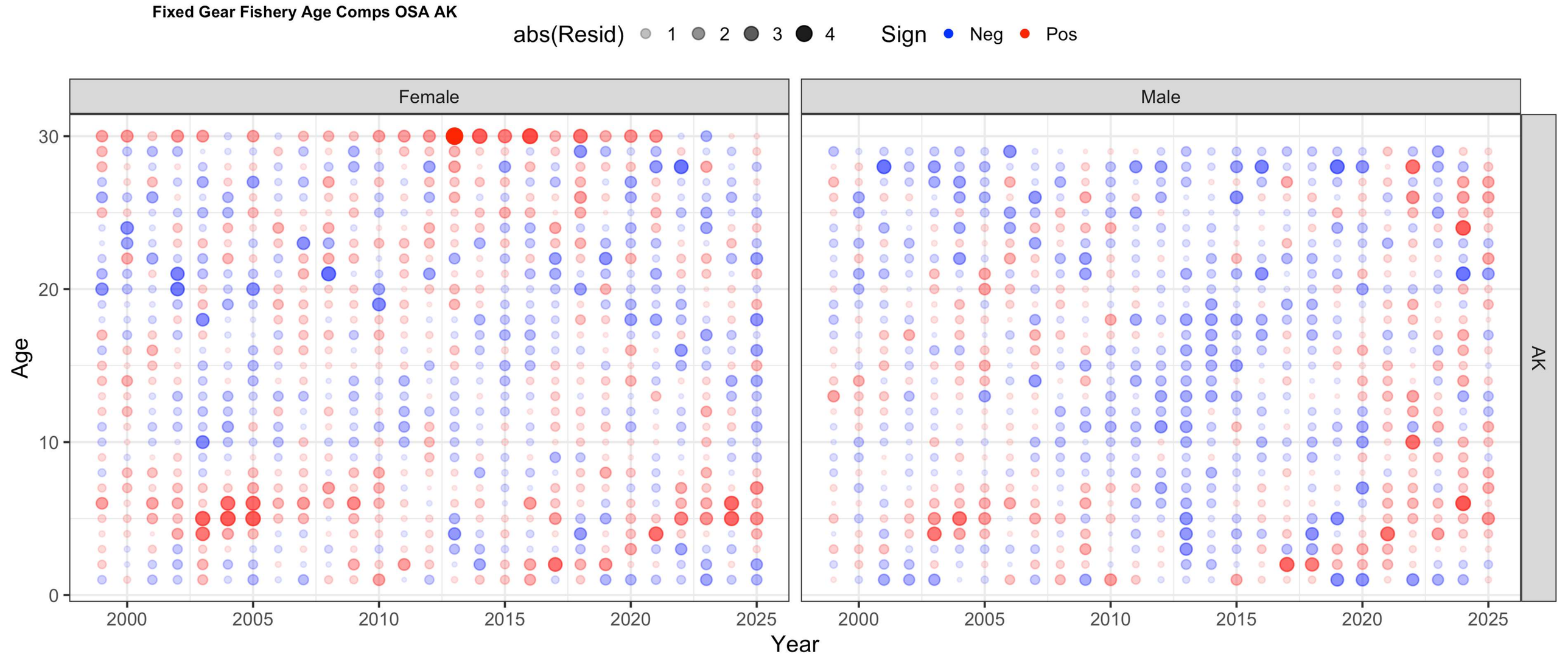
Fixed Gear Fishery Age Comps Female AK

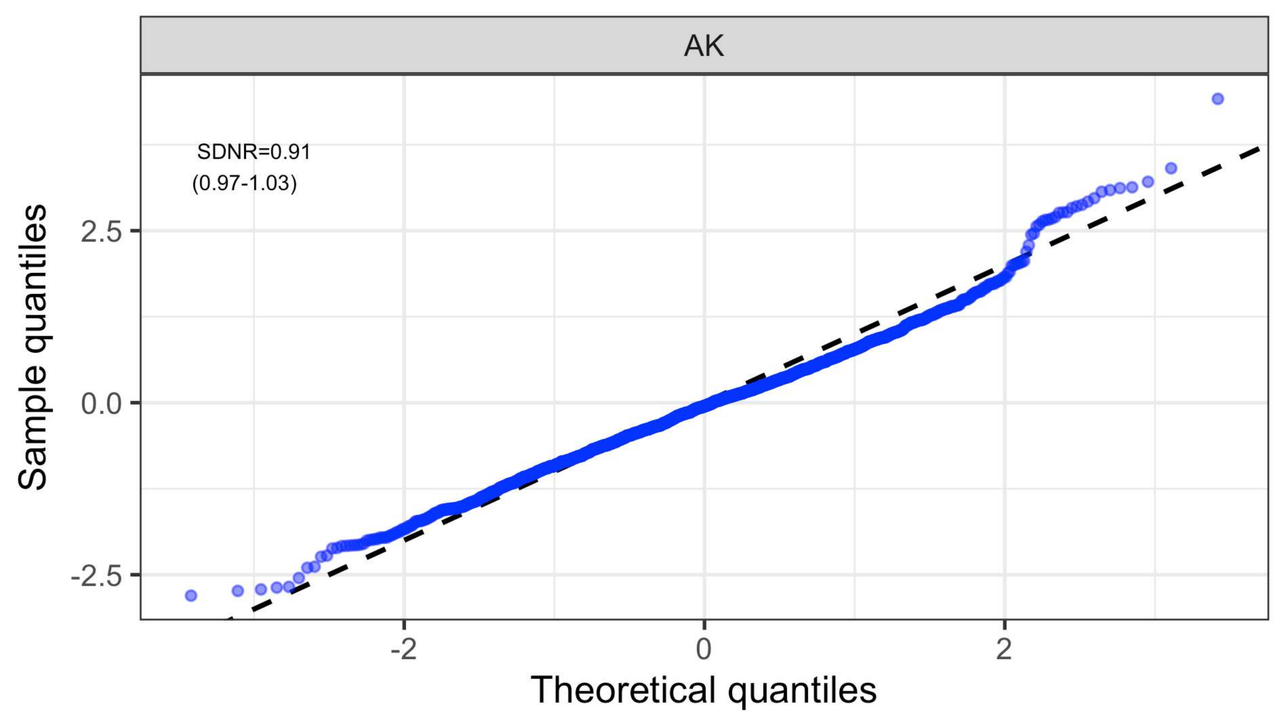


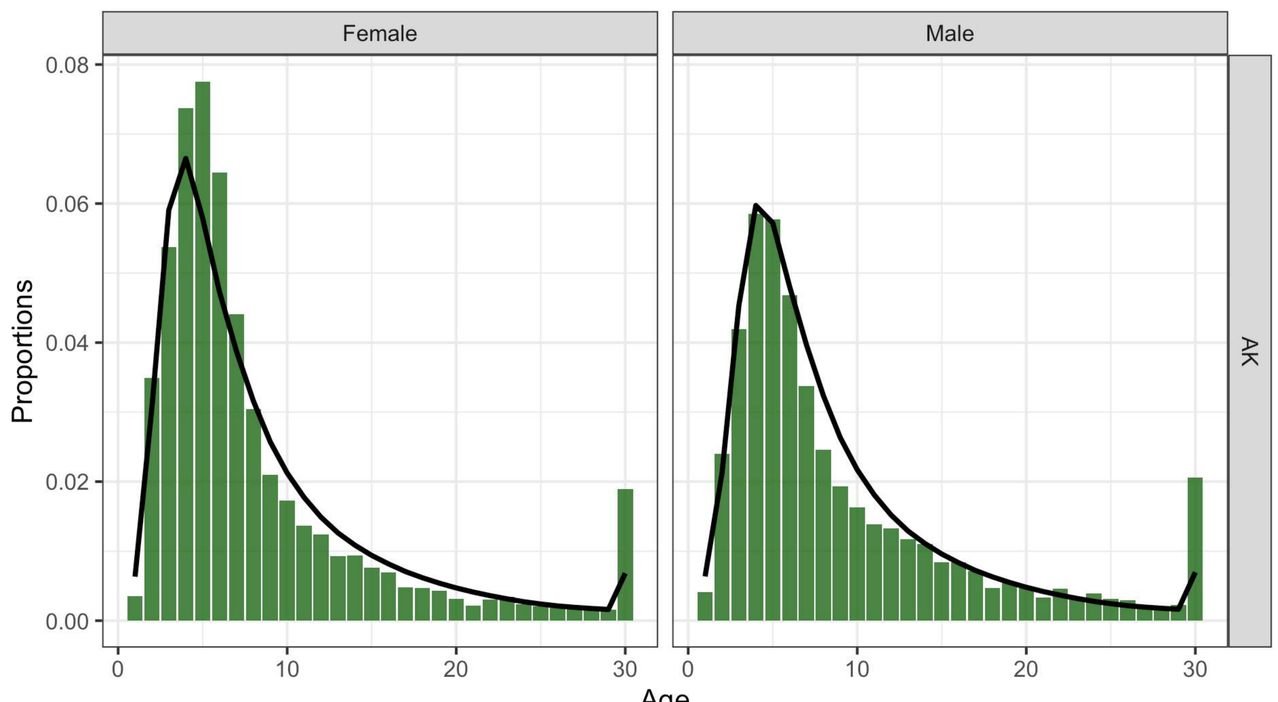
Fixed Gear Fishery Age Comps Male AK



Fixed Gear Fishery Age Comps OSA AK







Fixed Gear Fishery Age Comps Pearson residuals
Sign
Negative
Positive
|Pearson resid|
2
4
6
Female
Male
AK
Age
Year
0
10
20
30
2000
2005
2010
2015
2020
2025

Type: Age, Index, Length, Prior, Catch, jnLL, Penalty

26.2_FAA_TV_Selex_LLS

Likelihood

0
-1000
-2000
-3000

Catch nLL
Fishery Age nLL
Fishery Length nLL
Fishing Mortality Penalty
Initial Age Penalty
jnLL
M Prior
Recruitment Penalty
Selectivity Penalty
Survey Age nLL
Survey Index nLL
Survey Length nLL

Likelihood Component

Numbers-at-Age

Sex — Female ···· Male

Age- 1 | Age- 2 | Age- 3 | Age- 4 | Age- 5 | Age- 6

Age- 7 | Age- 8 | Age- 9 | Age- 10 | Age- 11 | Age- 12

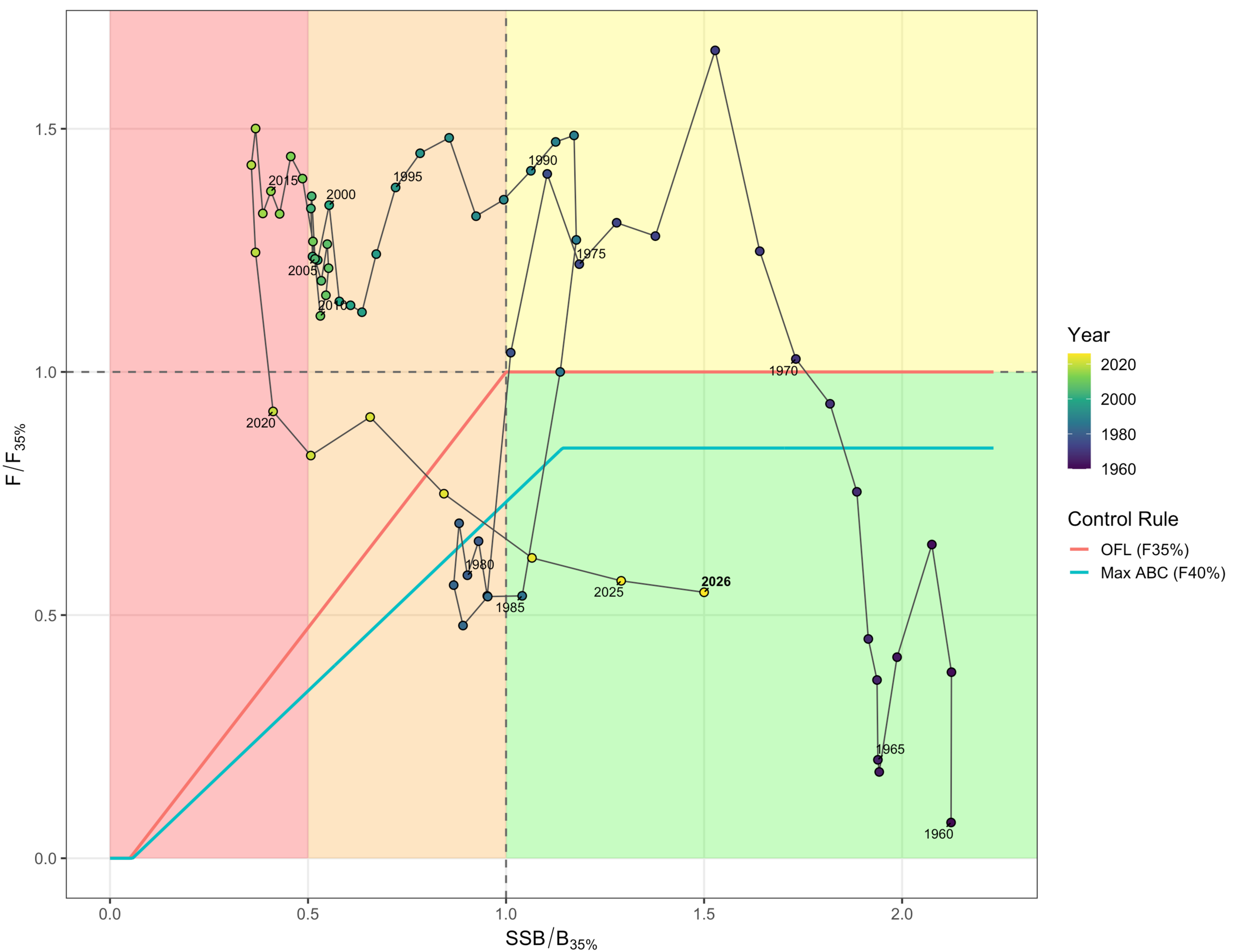
Age- 13 | Age- 14 | Age- 15 | Age- 16 | Age- 17 | Age- 18

Age- 19 | Age- 20 | Age- 21 | Age- 22 | Age- 23 | Age- 24

Age- 25 | Age- 26 | Age- 27 | Age- 28 | Age- 29 | Age- 30

Numbers-at-Age (millions)

Year


1.5
1.0
0.5
0.0
F/F35%
0.0
0.5
1.0
1.5
2.0
SSB/B35%
2020
2015
2000
1995
1990
1975
1970
2005
2010
1980
1985
2025
2026
1965
1960
Year
2020
2000
1980
1960
Control Rule
OFL (F35%)
Max ABC (F40%)

Retrospective Year
1 2 3 4 5 6 7

Recruitment

Mohns Rho: -0.3537

SSB

Mohns Rho: 0.2725

AK

Relative Difference from Terminal Year

3
2
1
0
-1

1960 1980 2000 2020

1960 1980 2000 2020

Year

Peel
1 2 3 4 5 6 7

AK
Recruitment

AK
SSB

Value

100
50
0

300
200
100
0

1960
1980
2000
2020

Year

AK

Recruitment (millions)

100

50

0

1

2

3

4

5

6

7

Years since cohort was last estimated

Cohort

2016

2017

2018

2019

2020

2021

2022

2023

Retrospective Year
1 2 3 4 5 6 7

Recruitment

Mohns Rho: -0.3504

SSB

Mohns Rho: 0.2788

AK

Relative Difference from Terminal Year

8
6
4
2
0

1960 1980 2000 2020

1960 1980 2000 2020

Year

Peel
1 2 3 4 5 6 7

AK
Recruitment

AK
SSB

Value

Year

150
100
50
0

300
200
100
0

1960
1980
2000
2020

AK

Recruitment (millions)

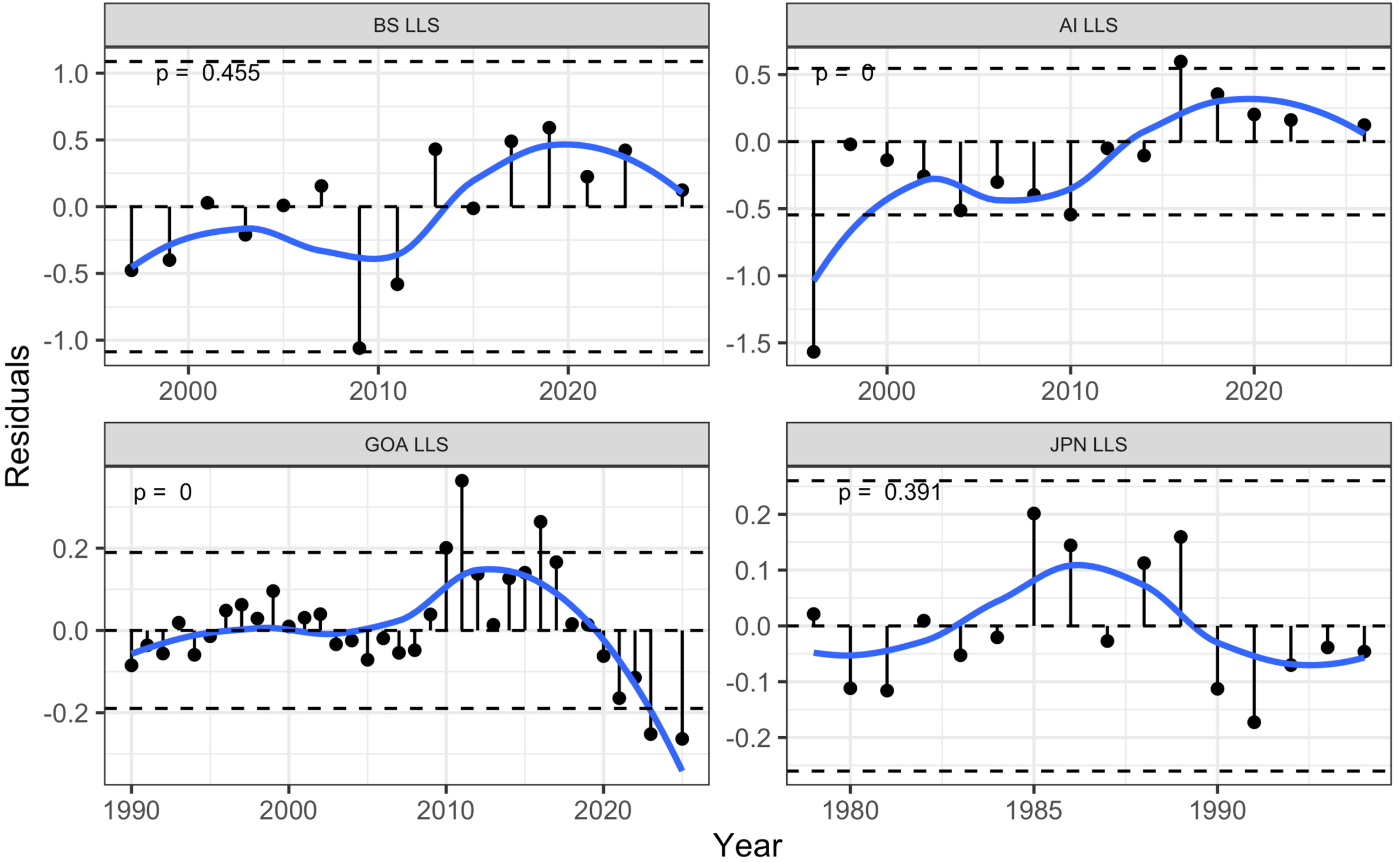
160
120
80
40

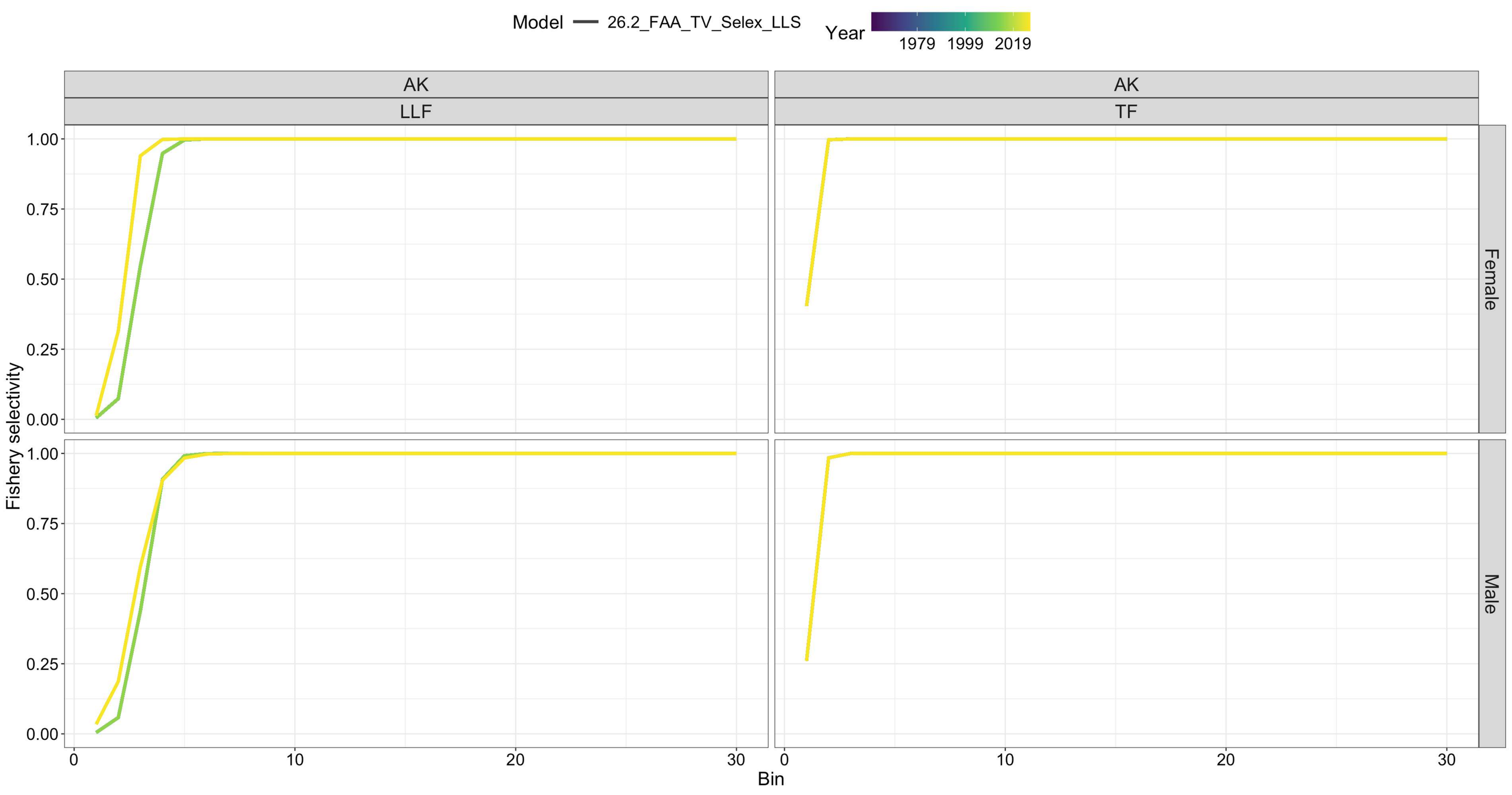
0
1
2
3
4
5
6
7

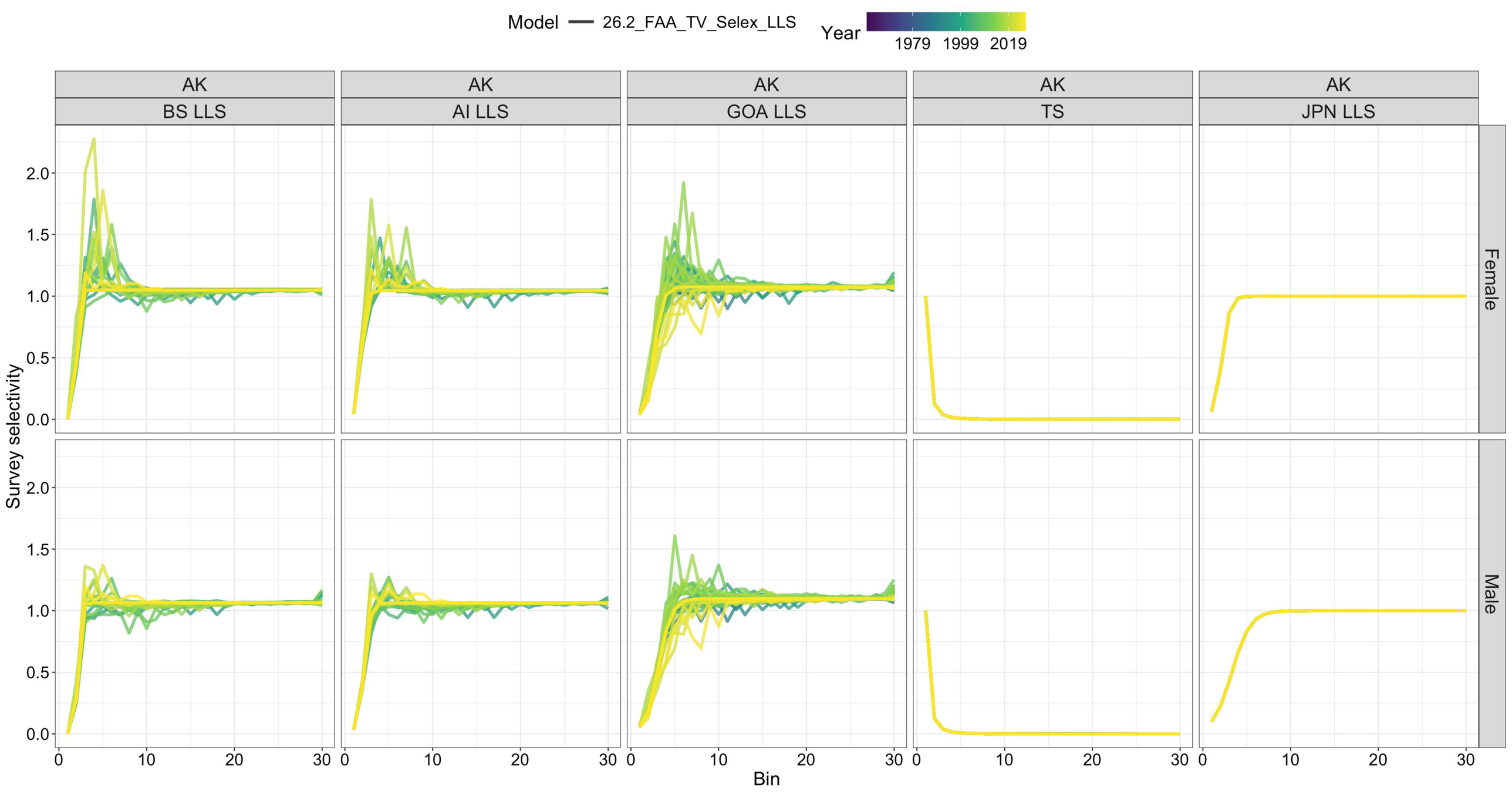
Years since cohort was last estimated

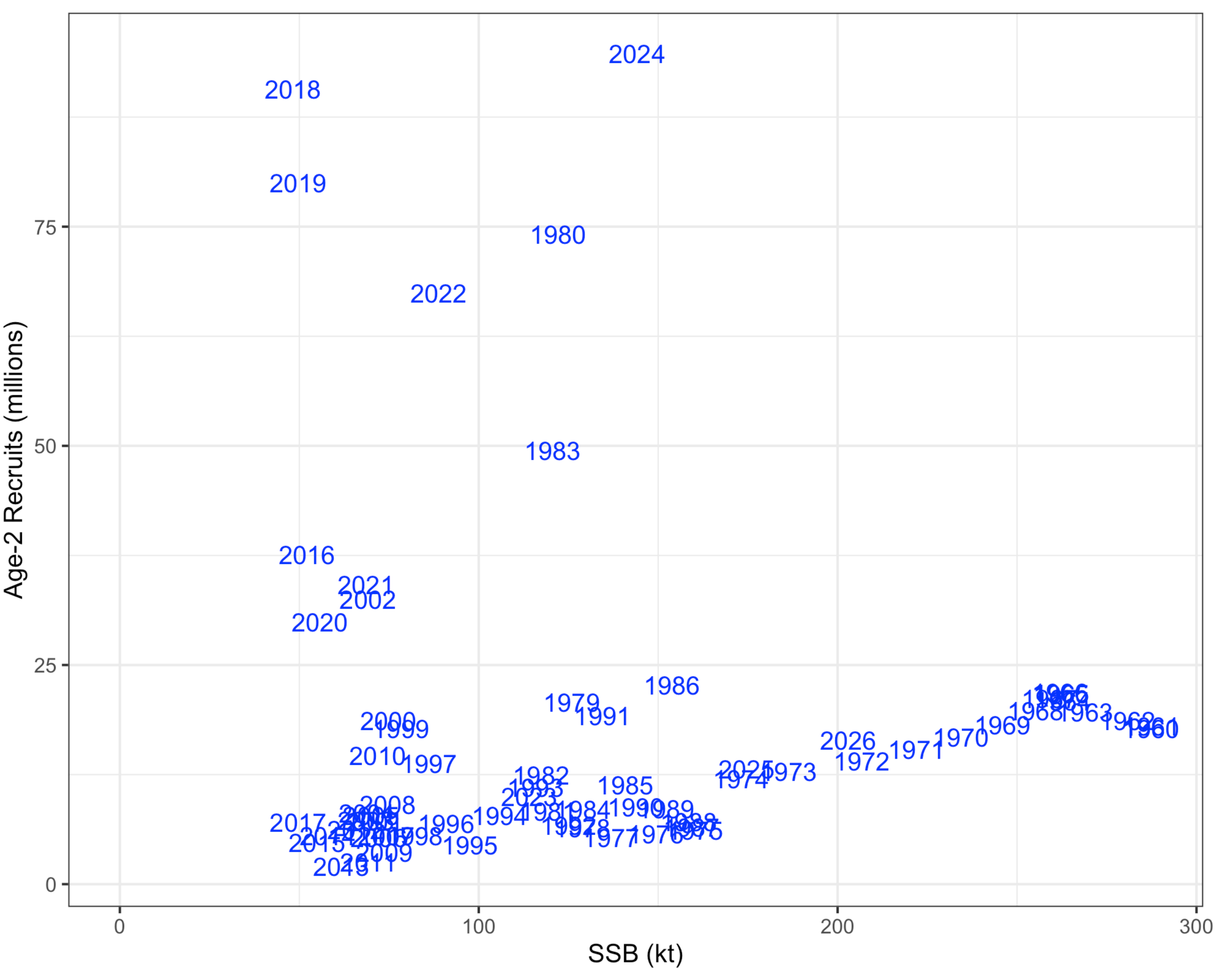
Cohort
2016
2017
2018
2019
2020
2021
2022
2023

Index Residuals Runs Test




Model
26.2_FAA_TV_Selex_LLS
Year
1979
1999
2019
AK
LLF
AK
TF
Female
Male
Fishery selectivity
1.00
0.75
0.50
0.25
0.00
0
10
20
30
Bin


Model
26.2_FAA_TV_Selex_LLS
Year
1979
1999
2019
AK
BS LLS
AK
AI LLS
AK
GOA LLS
AK
TS
AK
JPN LLS
Female
Male
Survey selectivity
0.0
0.5
1.0
1.5
2.0
0
10
20
30
Bin


Age-2 Recruits (millions)
0
25
50
75
0
100
200
300
SSB (kt)
2024
2018
2019
1980
2022
1983
2016
2021
2002
2020
1986
1979
1991
1999
2000
2010
1997
2026
1972
1971
1970
1969
1973
1985
1982
1995
1996
2017
2009
2011

Population and Catch Trends

100
75
50
25
0

Recruitment (millions of fish)

300
200
100
0

SSB or Catch (kt)

1960
1980
2000
2020

Year

Catch
Recruitment
SSB

Model 26.2_FAA_TV_Selex_LLS

AK

Dynamic SSB0

Recruitment

SSB

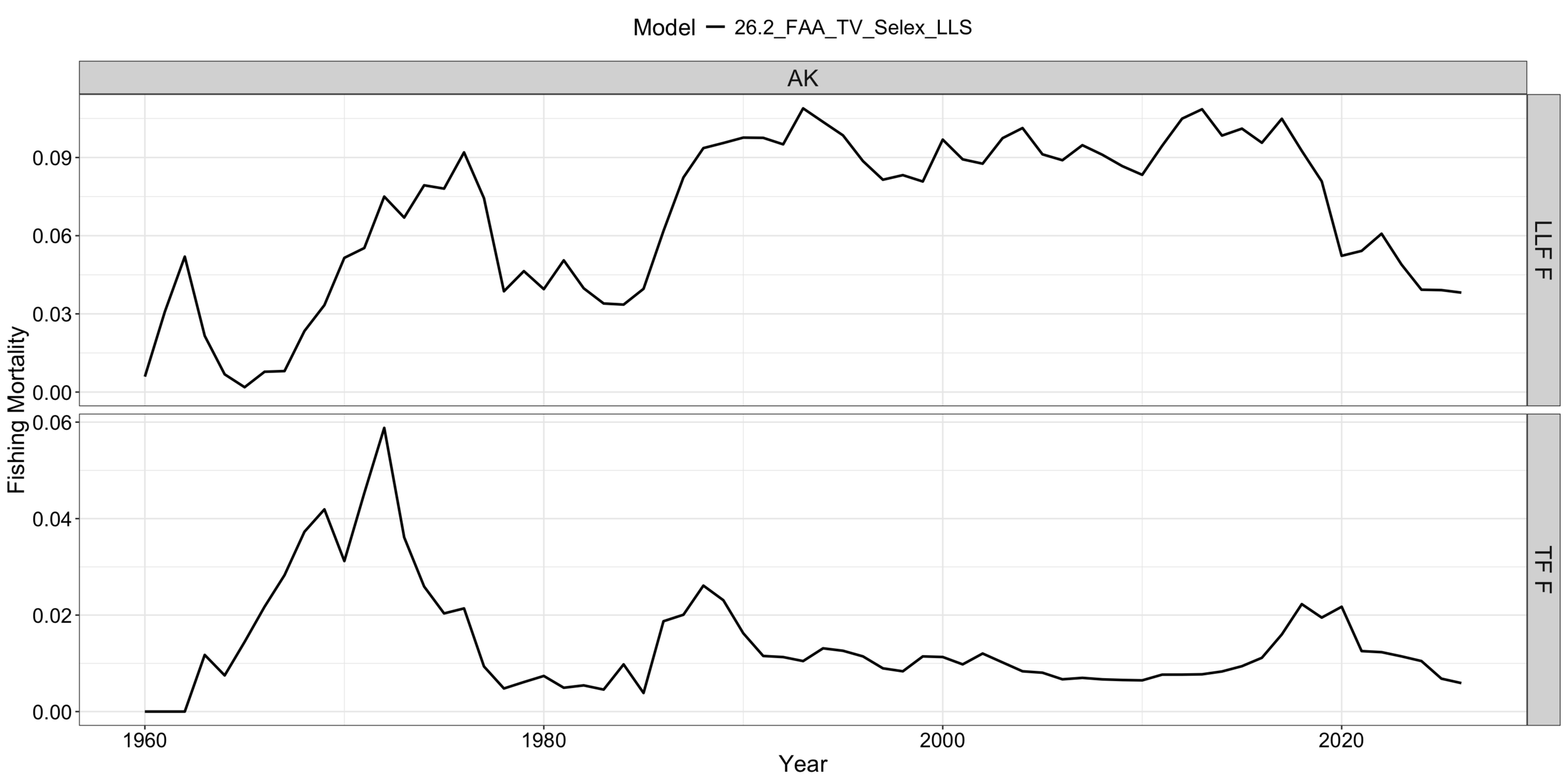
Total Biom

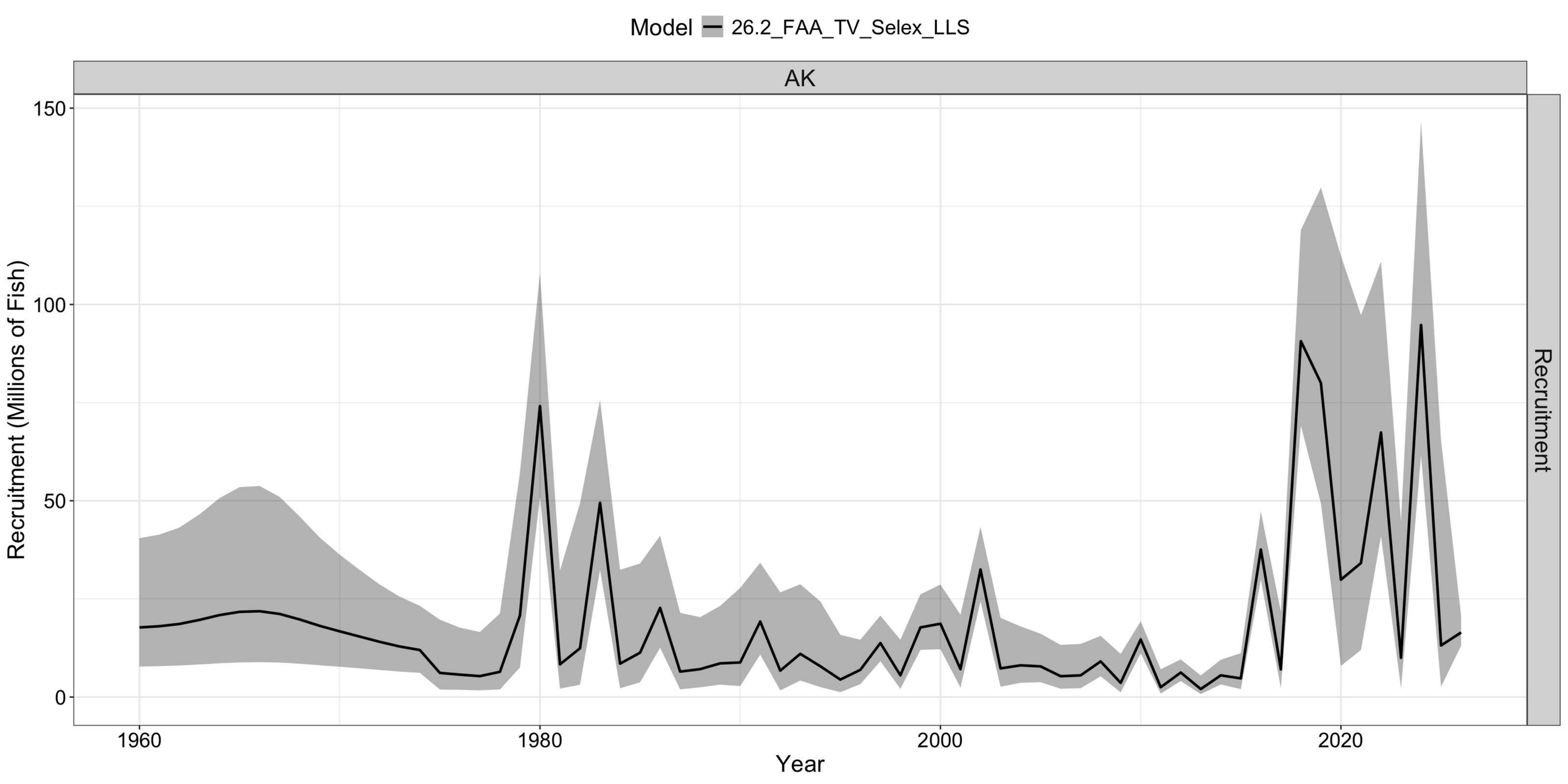
LLF F

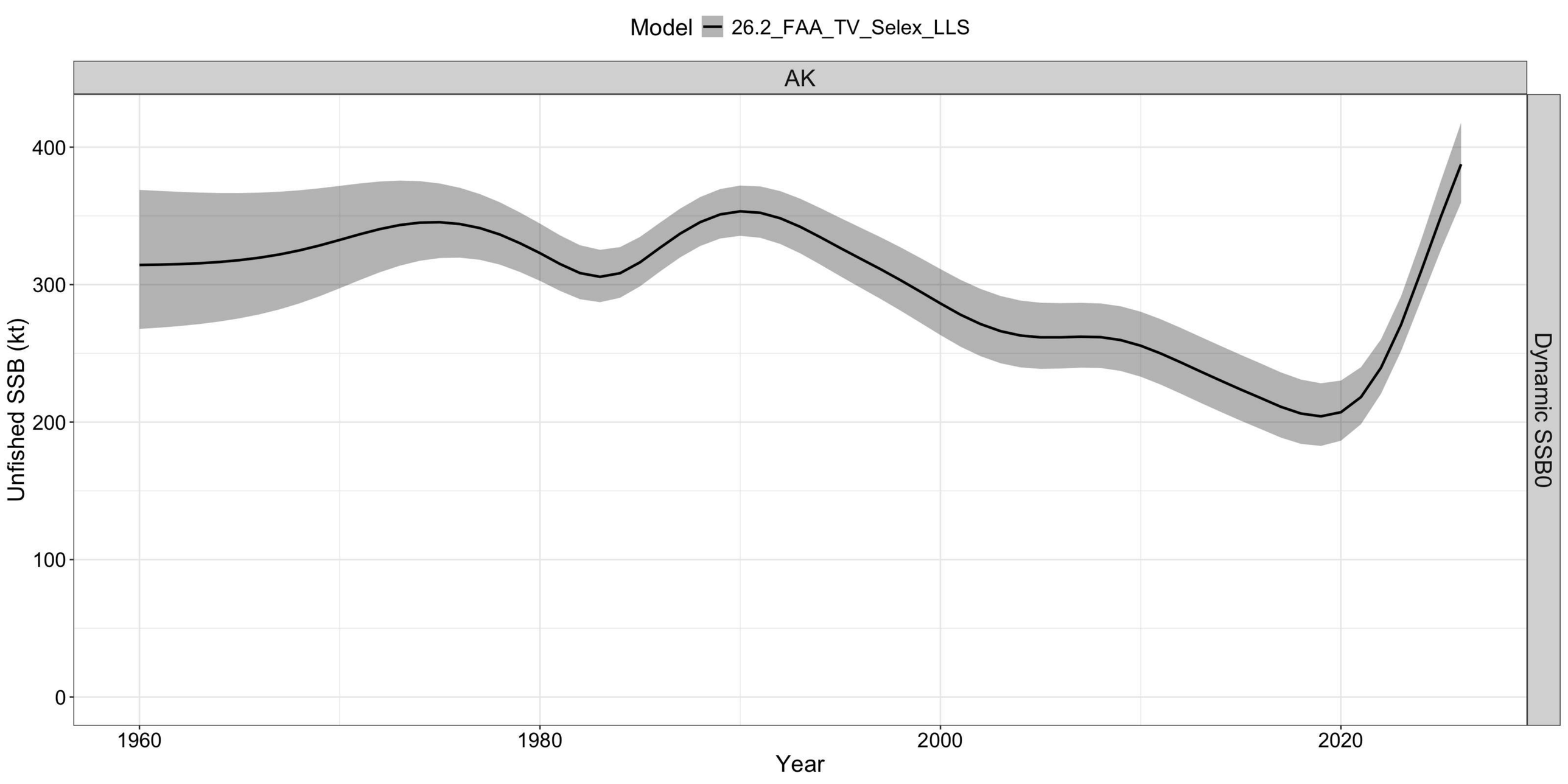
TF F

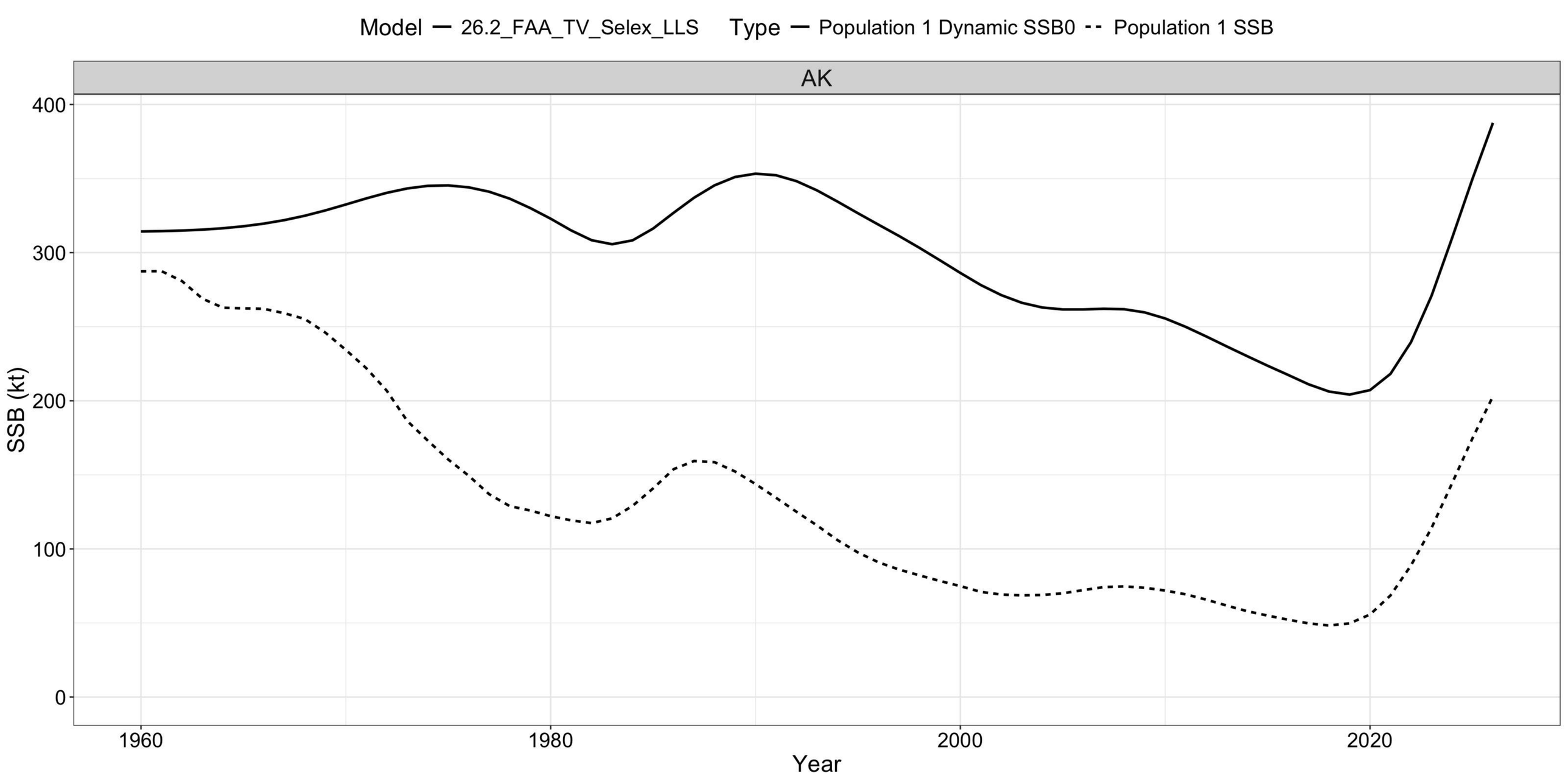
Value

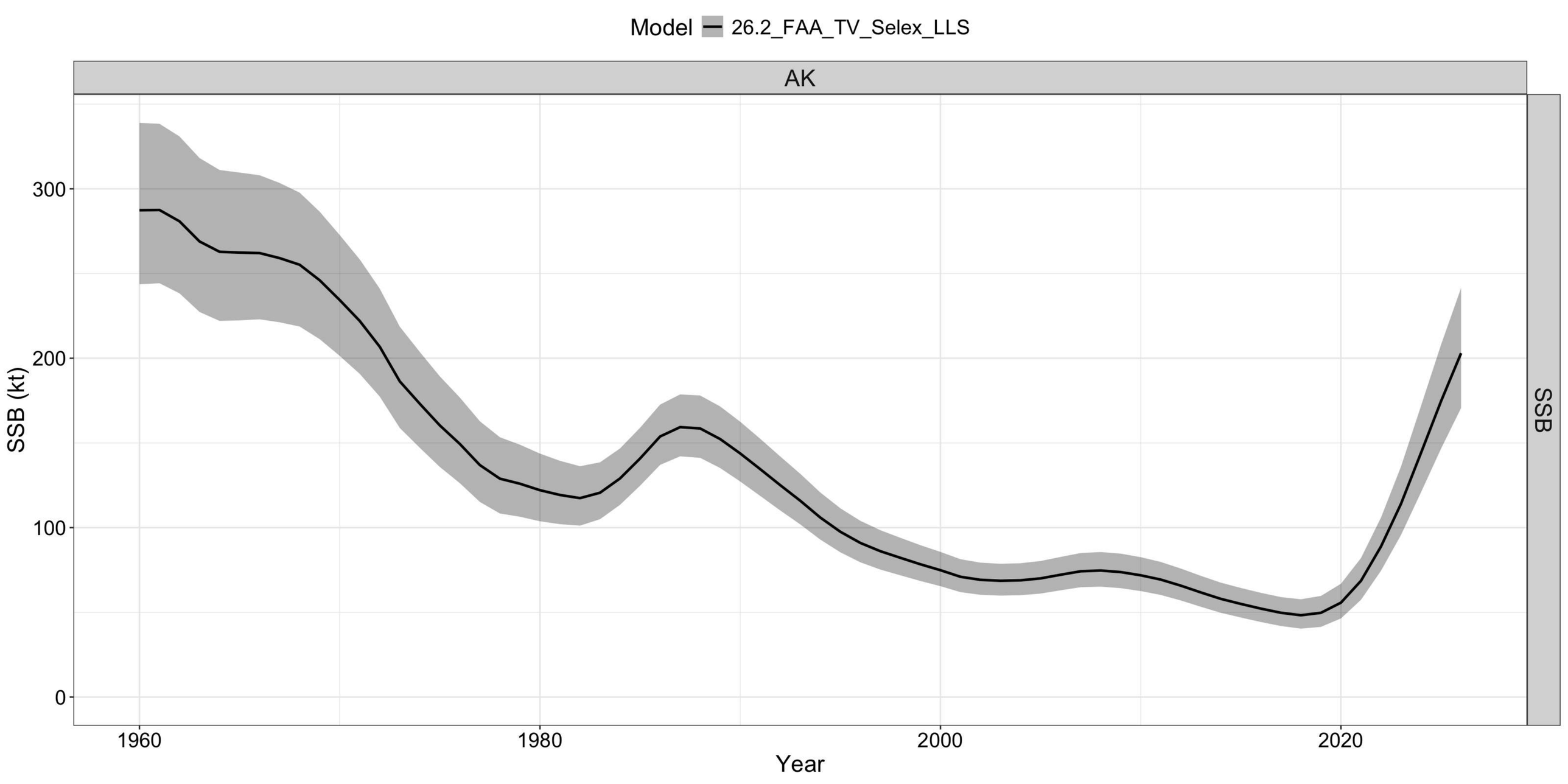
Year

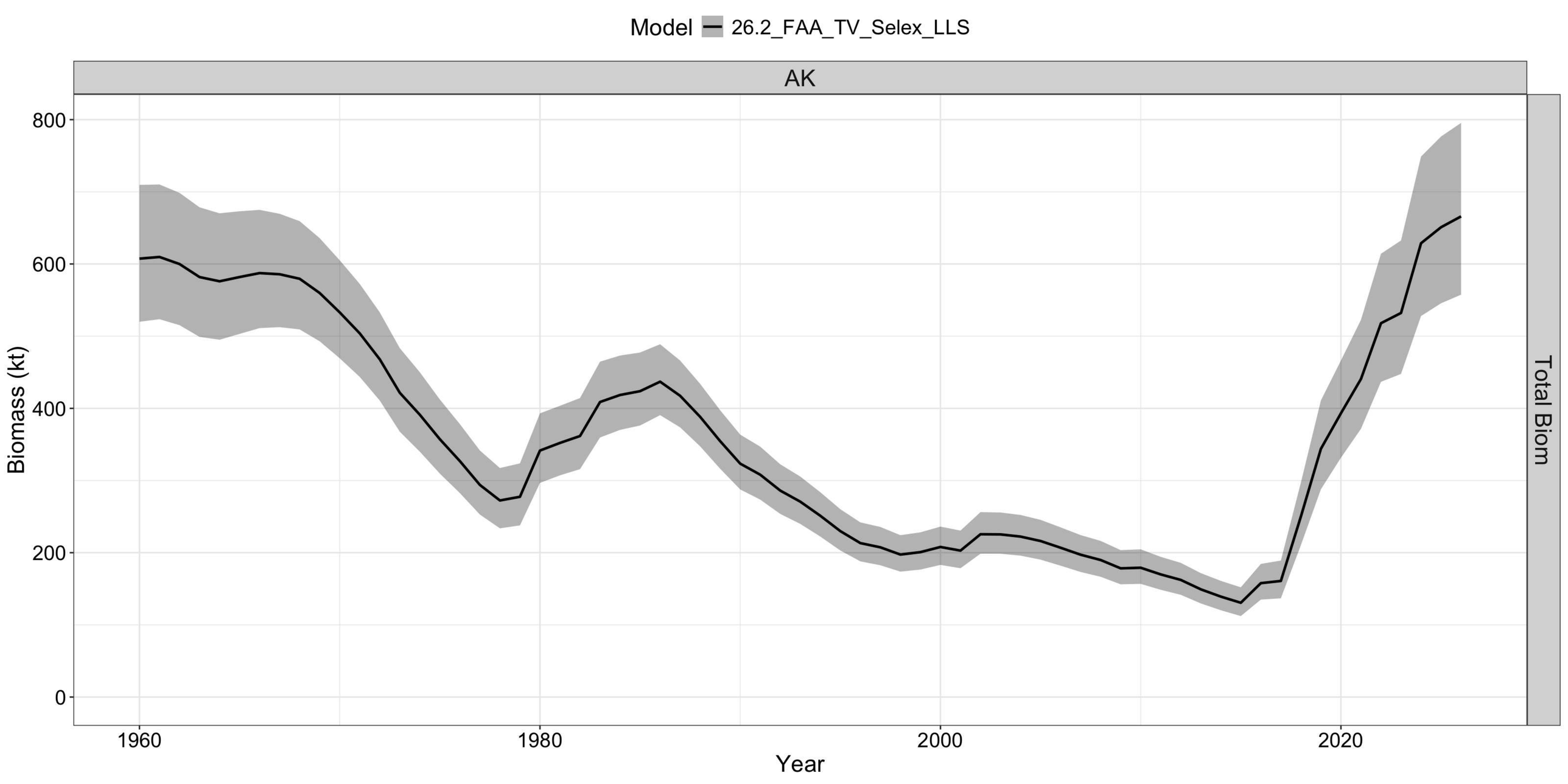

Model — 26.2_FAA_TV_Selex_LLS
AK
LLF F
TF F
Fishing Mortality
0.00
0.03
0.06
0.09
0.00
0.02
0.04
0.06
1960
1980
2000
2020
Year


Model
26.2_FAA_TV_Selex_LLS
AK
Recruitment
Recruitment (Millions of Fish)
150
100
50
0
1960
1980
2000
2020
Year

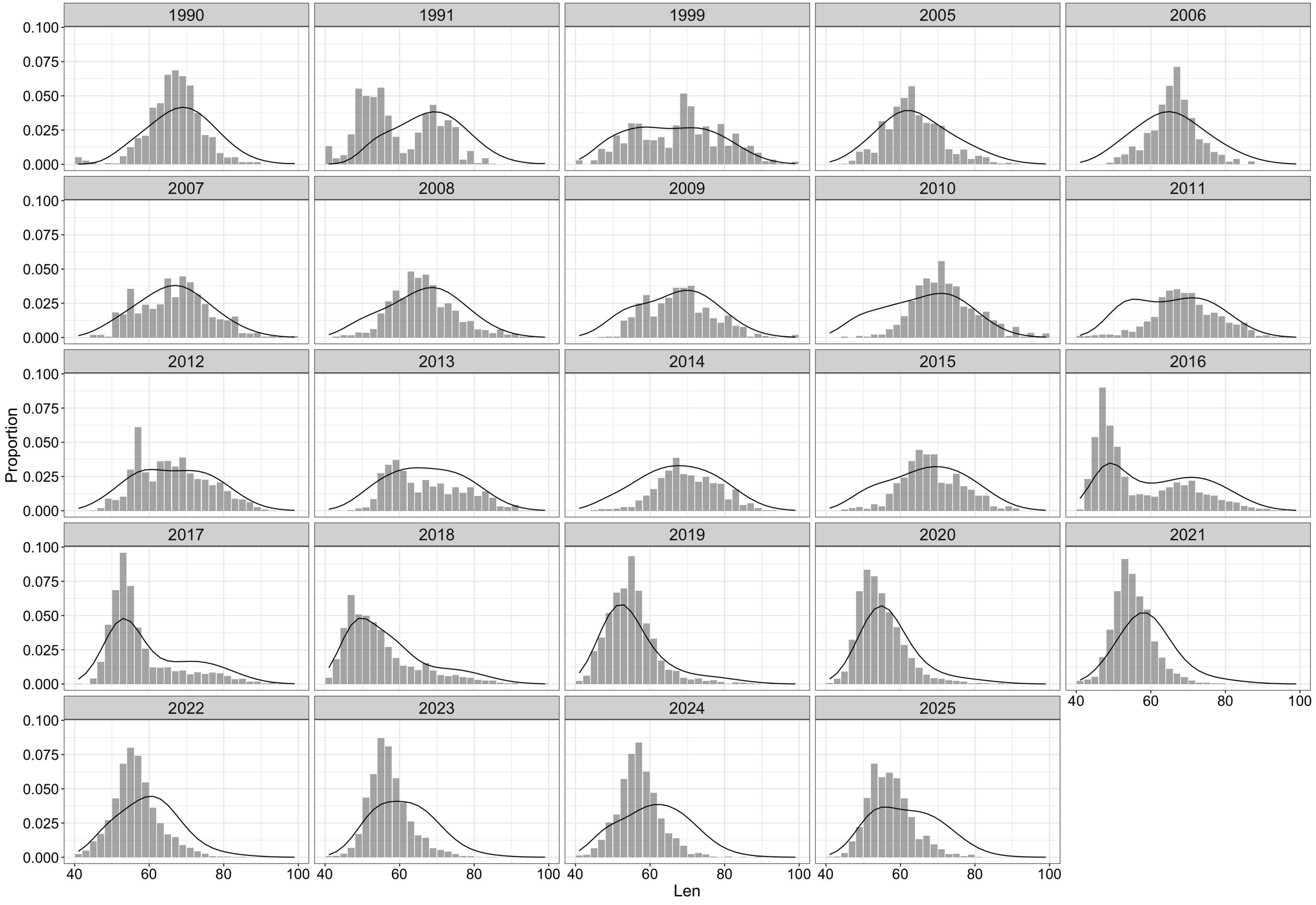

Model
26.2_FAA_TV_Selex_LLS
AK
Dynamic SSB0
Unfished SSB (kt)
400
300
200
100
0
1960
1980
2000
2020
Year


Model — 26.2_FAA_TV_Selex_LLS
Type — Population 1 Dynamic SSB0 -- Population 1 SSB
AK
400
300
200
100
0
SSB (kt)
1960
1980
2000
2020
Year

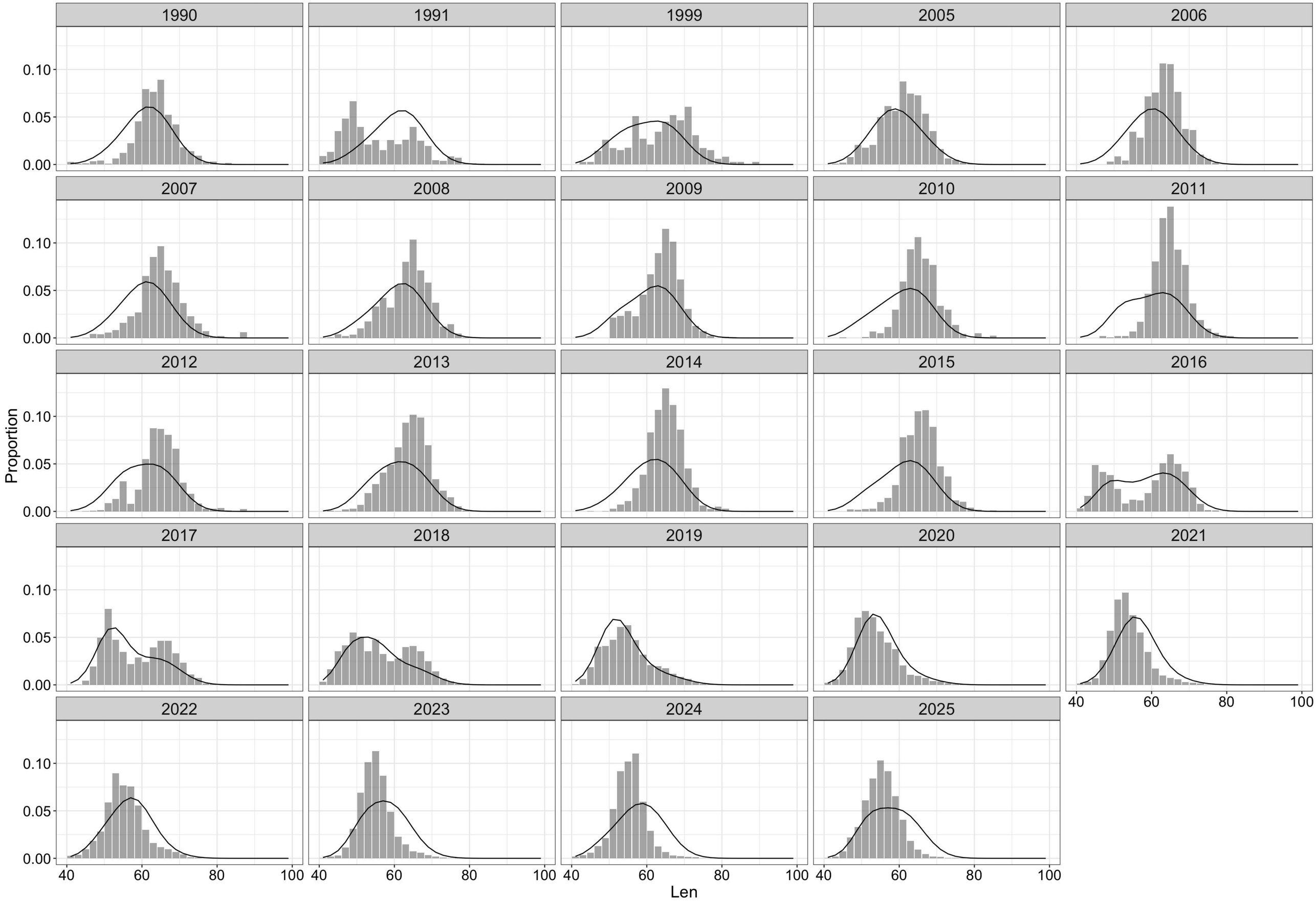

Model
26.2_FAA_TV_Selex_LLS
AK
SSB
SSB (kt)
300
200
100
0
1960
1980
2000
2020
Year


Model
26.2_FAA_TV_Selex_LLS
AK
Total Biom
Biomass (kt)
800
600
400
200
0
1960
1980
2000
2020
Year

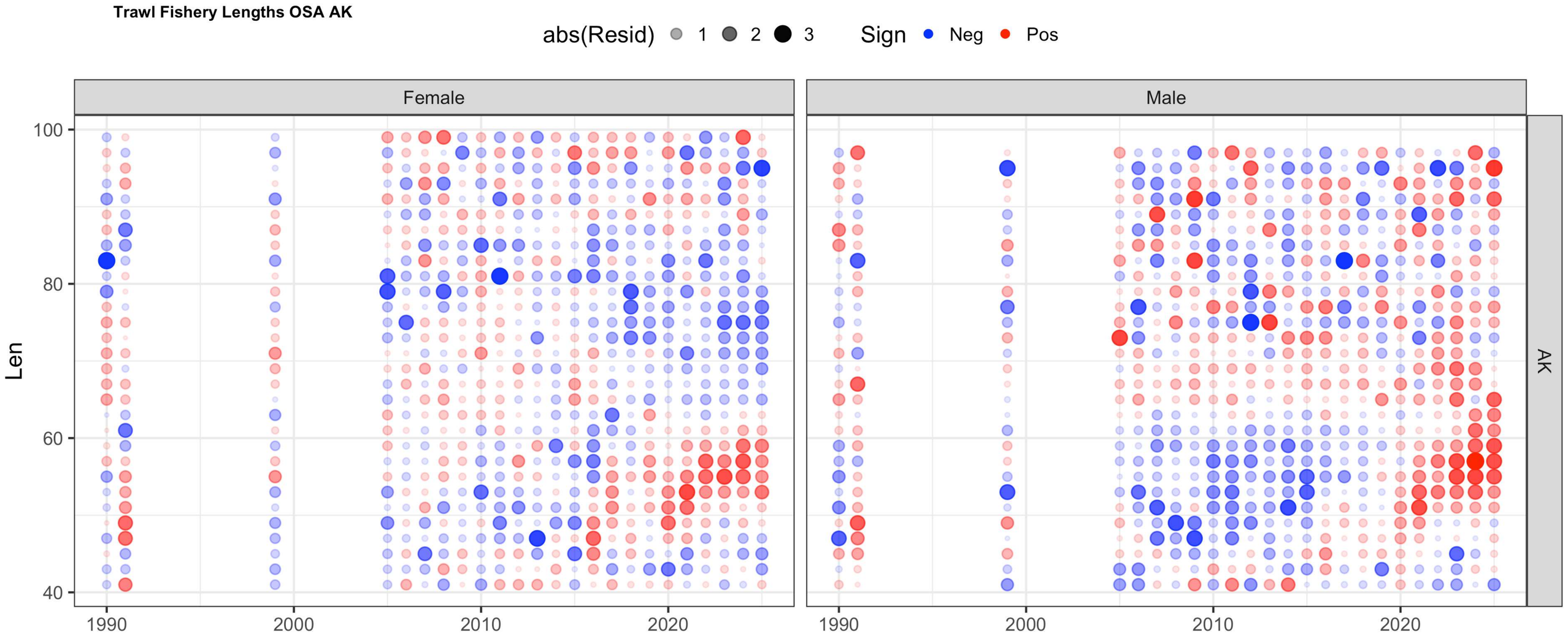
Trawl Fishery Lengths Female AK

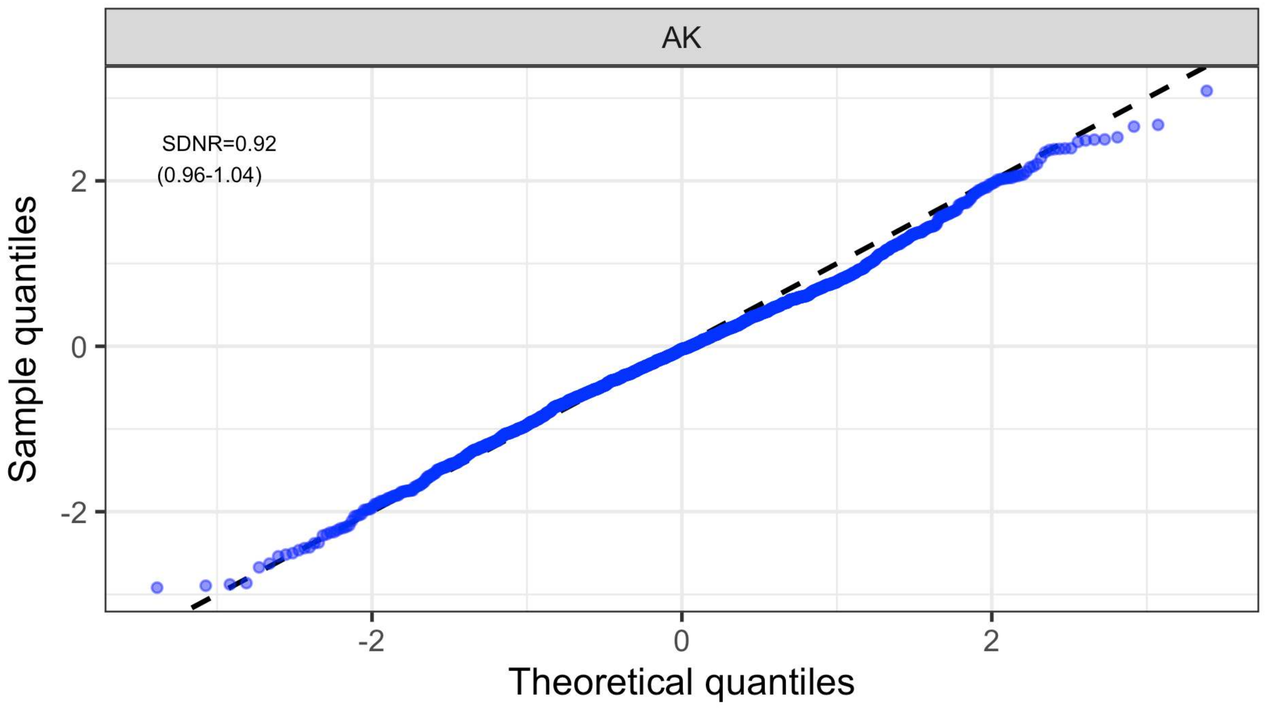


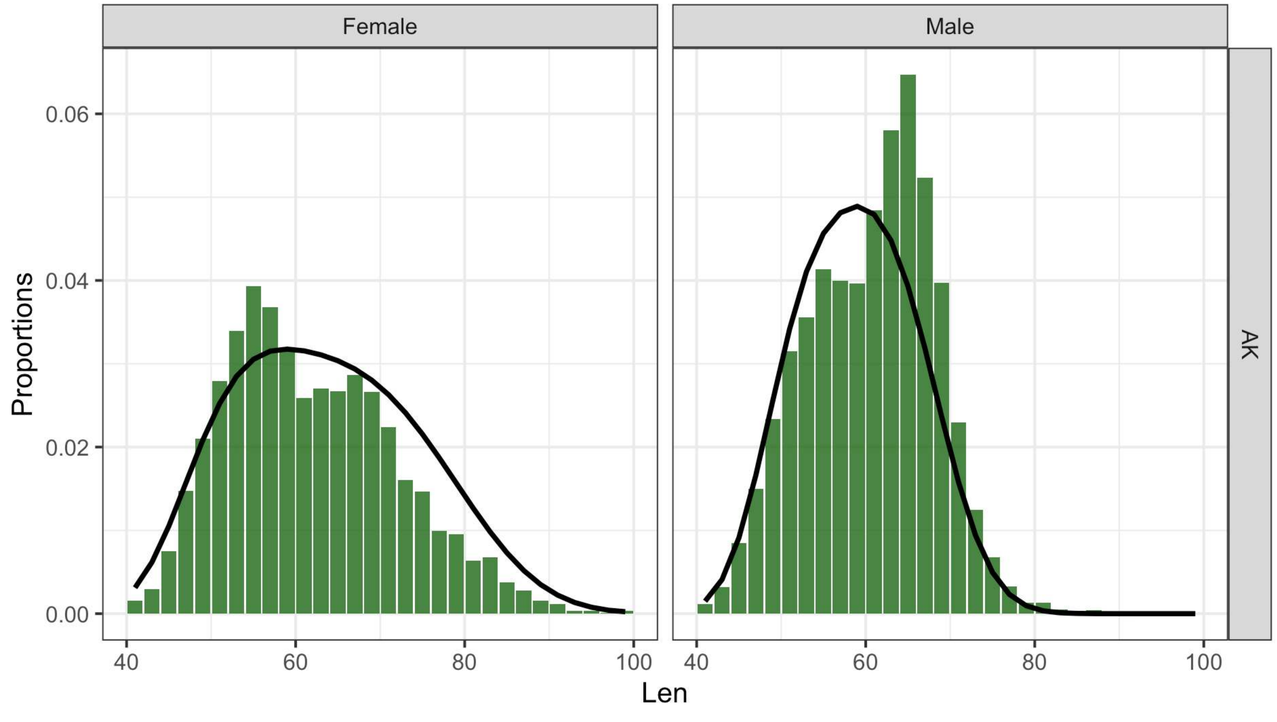
Trawl Fishery Lengths Male AK

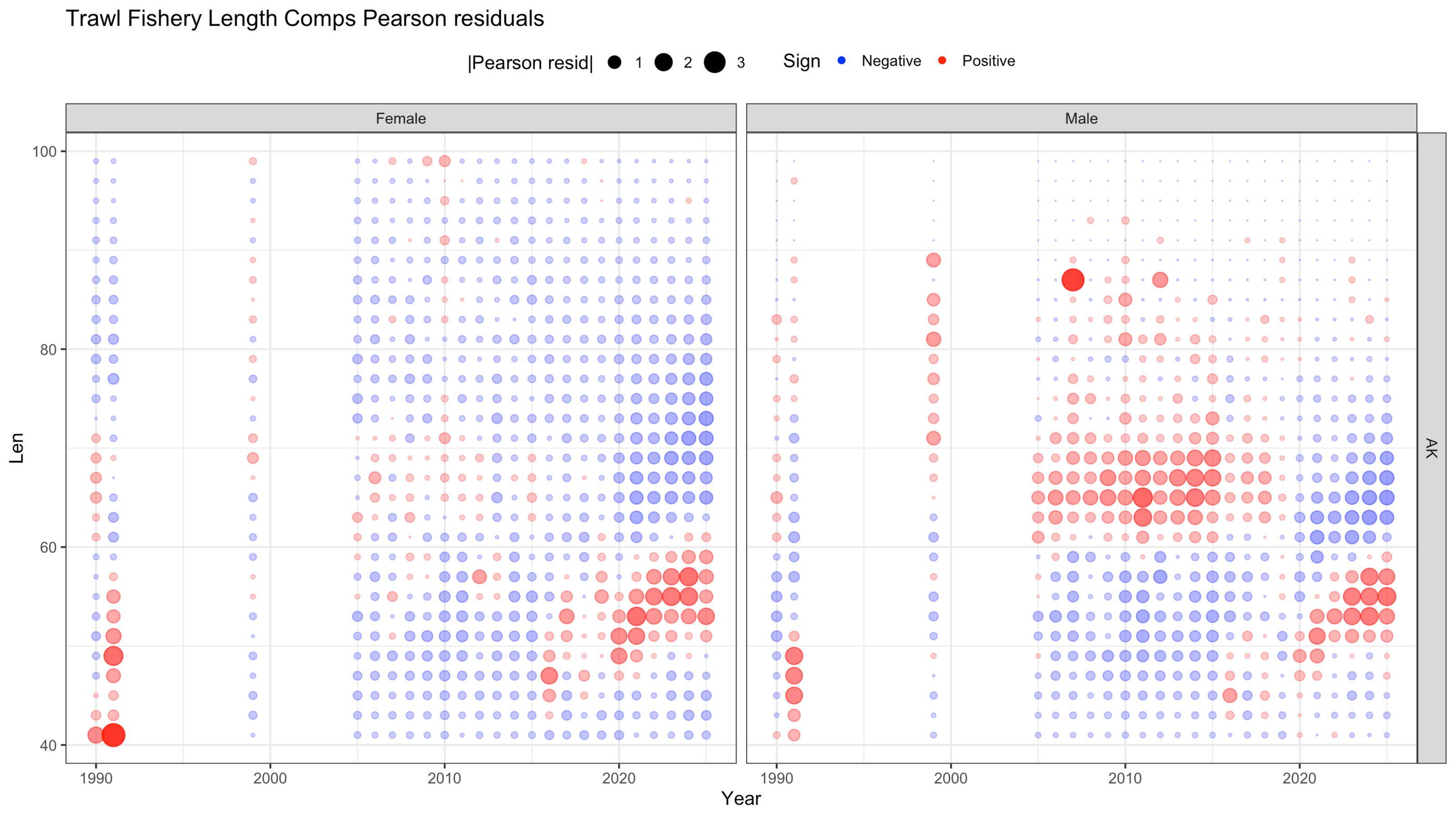


Trawl Fishery Lengths OSA AK

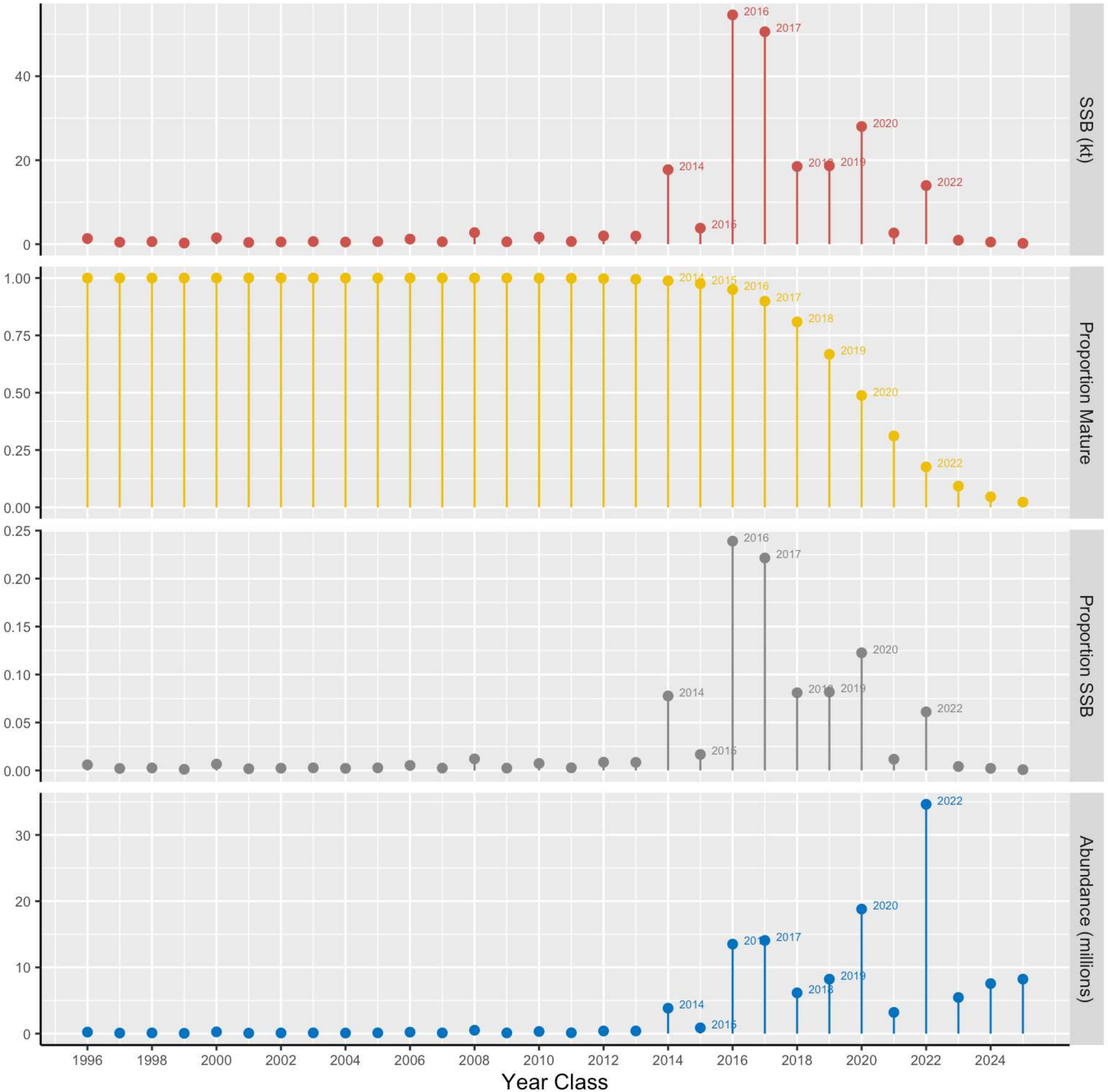






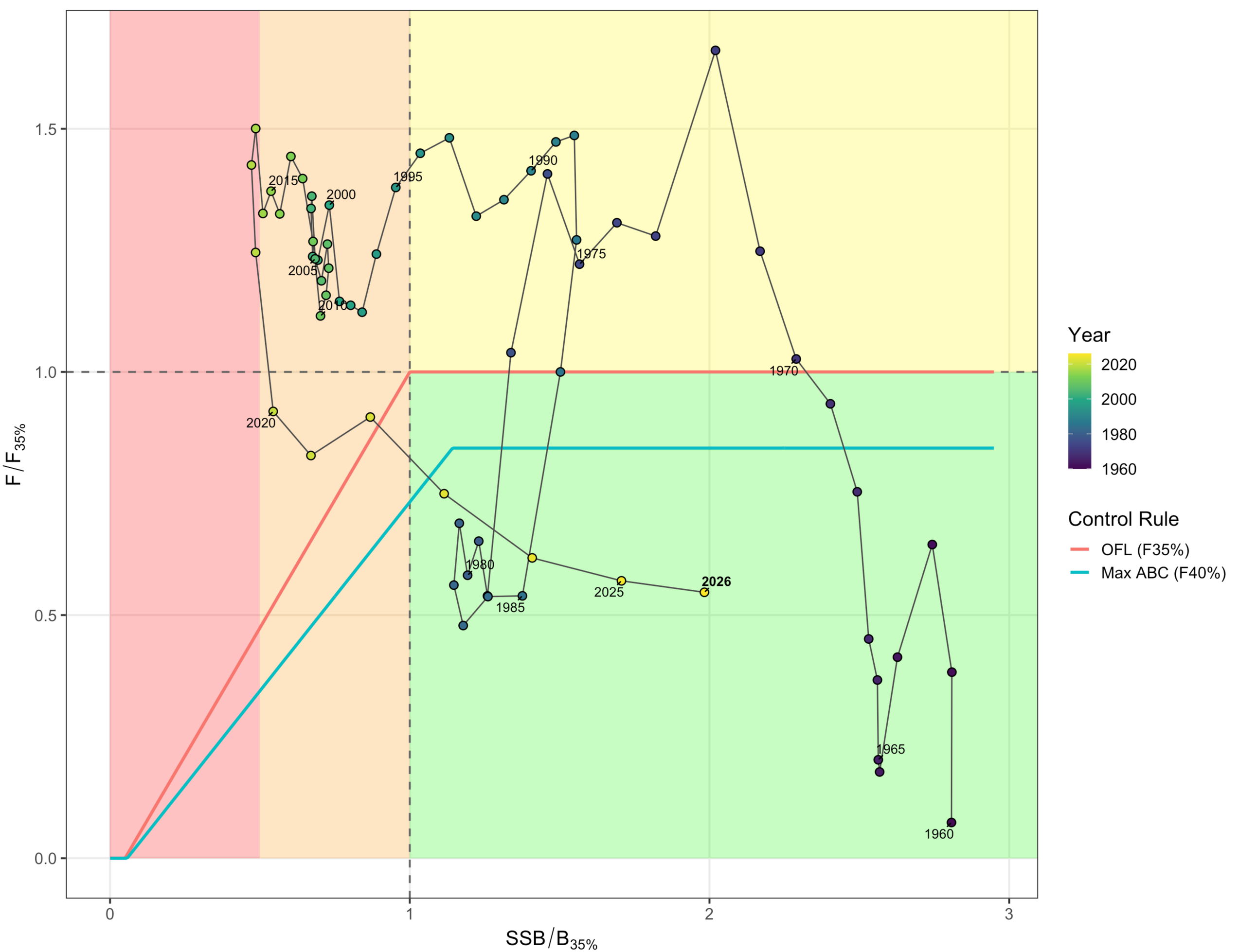

Trawl Fishery Length Comps Pearson residuals
|Pearson resid| 1 2 3
Sign Negative Positive
Female
Male
AK
Len
100
80
60
40
1990
2000
2010
2020
Year

Contribution to 2027 SSB by Year Class



Source	SE	Data_Wt	ISS	ESS
BS LL Survey Index	0.20	1	NA	NA
AI LL Survey Index	0.23	1	NA	NA
GOA LL Survey Index	0.08	1	NA	NA
Trawl Survey Index	NA	NA	NA	NA
JPN LL Survey Index	0.07	1	NA	NA
Fixed Gear Fishery CPUE	NA	NA	NA	NA
BS LL Survey Ages	NA	1	34.07	34.07
AI LL Survey Ages	NA	1	35.57	35.57
GOA LL Survey Ages	NA	1	67.17	67.17
JPN LL Survey Ages	NA	1	83.33	83.33
JPN LL Survey Lengths	NA	1	54.62	54.62
Fixed Gear Fishery Ages	NA	1	142.85	142.85
Trawl Fishery Lengths	NA	1	47.62	47.62

Model	Region	Terminal_SSB	Terminal_SSB0	Terminal_F	Catch_Advice	B_Ref_Pt	F_Ref_Pt	B_over_B_Ref	B_over_DynB_Ref	F_over_F_
2_FAA_TV_Selex_LLS	1	202.9844	387.5893	0.04405208	39.83523	154.6508	0.06795	1.31253	0.52371	0.64834


1.5
1.0
0.5
0.0
0
1
2
3
F/F35%
SSB/B35%
2020
2015
2000
1995
1990
1975
1970
2005
2010
1980
1985
2025
2026
1965
1960
Year
2020
2000
1980
1960
Control Rule
OFL (F35%)
Max ABC (F40%)

Quantity	2025	2026	2027	2028
M Female (estimated)	0.114	0.114	0.083	0.083
M Male (estimated)	0.114	0.114	0.082	0.082
Average Recruitment (millions)	25.9	25.9	20.1	20.1
Tier	3a	3a	3a	3a
Projected Total Biomass (Age-2+, t)	704,713	695,681	673485	679248
Projected Female Spawning Biomass (t)	219,714	241,217	228476	252608
B100% (t)	302,672	302,672	386627	386627
B40% (t)	121,069	121,069	154651	154651
B35% (t)	105,935	105,935	135319	135319
FOFL	0.102	0.102	0.081	0.081
max FABC	0.087	0.087	0.068	0.068
Total Depredation (t)	171	169	131	131
OFL (t)	58,731	57,993	46938	47357
OFLw (t)	58,532	57,797	46807	47226
max ABC (t)	50,283	49,651	39830	40187
ABC (t)	50,283	49,651	39830	40187
ABCw (t)	50,111	49,482	39699	40056
Status	2023	2024	2025	2026
Overfishing	No	NA	No	NA
Overfished	NA	No	NA	No
Approaching Overfished	NA	No	NA	No

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Fishing Mortality

0.00

0.05

0.10

1960

1980

2000

2020

2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Fishing Mortality

0.125
0.100
0.075
0.050
0.025
0.000

2010
2020
2030
2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Biomass (kt)

0
300
600
900

1960
1980
2000
2020
2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Biomass (kt)

0

300

600

900

2010

2020

2030

2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Yield (kt)

0

20

40

1960

1980

2000

2020

2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Yield (kt)

0
10
20
30
40

1960
1980
2000
2020
2040

Year

Fleet
LLF
TF

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Yield (kt)

0
10
20
30
40

2010
2020
2030
2040

Year

Fleet
LLF
TF

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Yield (kt)

0
10
20
30
40
50

2010
2020
2030
2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Recruitment (millions)

0
25
50
75

1960
1980
2000
2020
2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Recruitment (millions)

0

25

50

75

2010

2020

2030

2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

SSB (kt)

0

100

200

300

400

500

1960

1980

2000

2020

2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

SSB (kt)

0

100

200

300

400

500

2010

2020

2030

2040

Year